Beaches or Books:
Tourism and Human Capital Accumulation*

Matthew McKetty

University of Wisconsin – Madison

Abstract

Schooling decisions by households are shaped by the opportunity costs of lost wages inherent in children pursuing an education. Research has shown for example, that positive manufacturing shocks in an area can increase drop-out rates. What is still not well understood, is the extent to which shocks to demand for major service sector industries influence household educational investments. I fill this gap in the literature by studying the impacts of increases in tourism intensity on schooling in the context of Jamaica, where tourism exports comprise a major share economic output. Exploiting temporal and geographic variation in tourism activity across the country, I investigate the extent to which increased tourism demand shocks interact with household socioeconomic status, parental schooling, and geographic distance from schooling facilities to influence household investment in education at key points in a child's educational career. From these findings I then draw conclusions about the potential trade-offs between tourism induced changes in human capital investments for households and tourism-driven economic development.

1 Introduction

Human capital formation is a crucial component of structural transformation and economic growth. Growth of a country's human capital is dependent in large part on household decisions about educational attainment and investments. It is therefore important to understand how economic factors shape investments in schooling, the manner in which the growth of specific sectors can influence these investments, and the degree of heterogeneity in in observed effects across socioeconomic status, and geography. The existing literature has studied how booms in export manufacturing Atkin (2016), agricultural productivity shocks (Shah and

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Steinberg 2017), and commodity price shocks (Carrillo 2020) among others, can have substantial impacts on education attainment. However, there has been a lack of scholarly work studying the ways in which shocks to low and medium-skilled services influence educational attainment in exposed areas. This paper addresses this gap in the literature by examining the effects of specialization in tourism services on schooling enrollment and investments in Jamaica.

This paper answers the question: Do higher levels of municipality tourism result in lower levels of educational attainment and investment for exposed children and youths? More broadly, this study examines the impact of specialization in low and medium-skilled services on educational outcomes in a small open economy. This question is especially relevant to a sizable group of lower and middle-income countries in region such as Sub-Saharan Africa and Latin America that are transitioning directly from economies based around natural resource extraction, to those focused on services (Nayyar et al. 2021). Theoretically, the effects of these transitions depend on two channels: an income effect, whereby higher earnings lower the marginal utility of consumption and encourage greater investment in schooling, and a substitution effect, whereby higher wages raise the opportunity cost of schooling. Influencing both of these channels are the expected returns to continuing education. Which effect dominates in cases of low and medium-skilled service sector specialization is the empirical question at the heart of this paper.

As an upper middle-income country which has invested considerably in the export-focused service of tourism, Jamaica is an excellent setting in which to study the effects of specialization in low and medium-skilled services on schooling. Like many emerging economies, over the past half-century Jamaica's economy has transitioned from being based upon natural resource (Bauxite and Agriculture), to one being based upon services such as tourism (King 2001). Tourism services comprise over 33 percent of economic activity when considering back-linkages (World Bank 2022), and earnings have grown by an average of 5 percent since 2000 (Jamaican Ministry of Tourism 2025). The country has however, struggled to maintain consistent GDP growth [1], and an underlying cause has been the lack of growth in

[1] Source: (World Bank 2022)

labor productivity.[2]

To carry out this analysis, I combine unique and granular tourist spending data with detailed schooling data from Jamaican household surveys for the years 2006, 2010, 2012, 2014, and 2018. Data on tourism activities comes from exit surveys collected by the Ministry of Tourism on a representative sample of visitors to the island. Data on educational attainment and household investments come from the Jamaica Survey of Living Conditions (JSLC), a nationally representative household survey managed by the Statistical Institute of Jamaica (STATIN). The JSLC provides information on enrollment, absenteeism, grade level, type of school attended, and education spending at the individual level across categories such as tuition, textbooks, and transportation. I link households and one-year lagged tourist expenditures across spatially consistent units, producing a repeated cross-section of over 13,000 school-aged children and young adults between the ages of 5 and 19. I use a shift-share instrumental variable strategy to account for the likely endogeneity of development-area tourism levels, leveraging plausibly exogenous variation in visits to Jamaica by tourists from different countries of origin.

The results show that higher one-year lagged area tourism levels produce no change in the likelihood that individuals aged 5 to 19 are enrolled in grade schools, nor any change in the relative popularity of different reasons provided for dropping out. Higher lagged tourism levels do reduce the likelihood of college-aged adults being enrolled in a university of any kind. I interpret this result as suggesting that higher tourism levels raise the opportunity cost of a college education. In terms of schooling attendance, I obtain mixed results, with higher prior-year tourism levels reducing frequent and occasional absenteeism resulting from illnesses, but also increasing absenteeism attributed to transportation costs. I argue that the first result indicates positive spillovers to education from higher earnings resulting from increased spending on healthcare services from tourism booms as was observed in McKetty (2026). Additionally, I argue that the increase in transportation-related absences reflects a crowding out of local consumption of transportation services by tourists, which I support by demonstrating that the effect is entirely driven by rural students, who in general must travel

[2] See: Caribbean National Weekly

further for schooling.

The findings of this study are relevant to research and policy on school enrollment, the intersection of health and human capital accumulation, and the spillover effects of local sector booms on educational attainment. Unlike in cases such as Atkin (2016), tourism does not reduce completion of secondary school, but it does limit progression to tertiary schooling. This may reflect the fact that tourism is largely composed of mid-skilled occupations, or that the major beneficiaries of increases in local tourism services are mid-skilled occupations (McKetty 2026). The finding that higher tourism is associated with lower illness-related absenteeism presents makes clear the importance of economic shocks not just for the choice to enroll in school, but for the ability to benefit from that enrollment. The result that tourism increases transportation-cost related absenteeism for rural students shows the potential for tourism induced crowding out (Allen et al. 2021) spillovers to exacerbate existing inequalities in access to education. Thus, this paper highlights understudied channels through which local sector booms impact educational outcomes.

This paper contributes to three areas of the literature, with the first being the expansive literature studying the factors influencing human capital formation. Some work in this area has focused on the role of the expansion of export-lead industrialization on schooling decisions (Nguyen 2025; Cai et al. 2024; Atkin 2016; Heath and Mobarak 2015), with studies such as those by Bau et al. (2020), Edmonds et al. (2010), and Edmonds and Pavcnik (2005) showing the important role played by child labor. Other studies have examined regions and countries specializing in agricultural commodities, and have used shocks to international prices (Beshir and Maystadt 2024; Carrillo 2020), or to weather (Zimmermann 2020; Shah and Steinberg 2017) as identifying variation. I contribute to this broad literature by illustrating the importance of channels such as health and access to transportation services on educational outcomes. Furthermore, this paper shows how shocks to mid-skilled services can both encourage completion of secondary schooling, while also reducing investment in tertiary education, an important mechanism for developing high-skilled labor.

This study also contributes to the literature studying services-lead structural transformation in open economies. One strand of this literature has shown the capacity of high-skilled services to drive productivity growth in contexts such as India (Peters et al. 2026; Fan et al. 2023). Rodrik (2016) first coined the term "premature de-industrialization" to describe the phenomenon of many emerging economies transitioning directly from natural resources to services, and described the uncertain implications of this divergent transition path. This paper contributes by studying the impacts of the growth of a major low and medium-skilled service sector on education decisions across all levels of schooling. This paper also highlights how the local consumption aspect of tourism services can crowd out local consumption and harm access to schooling as a result.

Finally, this paper also speaks to the "resource curse" literature which investigates the extent to which natural resource abundance can harm long-run development. Seminal work by Corden and Neary (1982) described how natural resource specialization can crowd out domestic manufacturing through currency appreciation, which they dubbed the "Dutch Disease". Other research in this space has focused on topics such as resource booms and human capital accumulation (Bonilla Mejía 2020), natural resources and backlinkages (Aragón and Rud 2013), and the importance of institutions for effective use of natural resources for development (Venables 2016). Natural endowments such as weather, beaches, and topography are crucial features for the 'Sun, Sand, and Sea' style of tourism that is common in Jamaica and other developing countries (Faber and Gaubert 2019; King 2001). Tourism sectors in developing contexts are also criticized for acting as enclaves that do not broadly benefit local populations. This paper highlighs a channel by which natural endowments monetized as a service, can impact human capital accumulation in developing country contexts.

The paper is organized as follows. Section Section 2 will provide an overview of the Jamaican tourism industry and education system. Section Section 3 provides an overview of the data used in the analysis. Section Section 4 discusses the empirical approach for the study. Section Section 5 details the results, and section Section 6 concludes.

2 Context and Background

2.1 Jamaica and Tourism

Jamaica is an upper-middle-income country in the northern Caribbean with an economy heavily based on services, which comprise approximately 70 percent of GDP. Tourism services account for 33 percent of GDP when factoring in backward linkages.[3] Even as tourism has grown considerably over the past two decades[4], the country has struggled to achieve meaningful economic growth, with average GDP per capita oscillating between 1 and -1 percent since 2000. A commonly cited reason is Jamaica's persistent challenge in expanding the skills and productivity of its workforce.

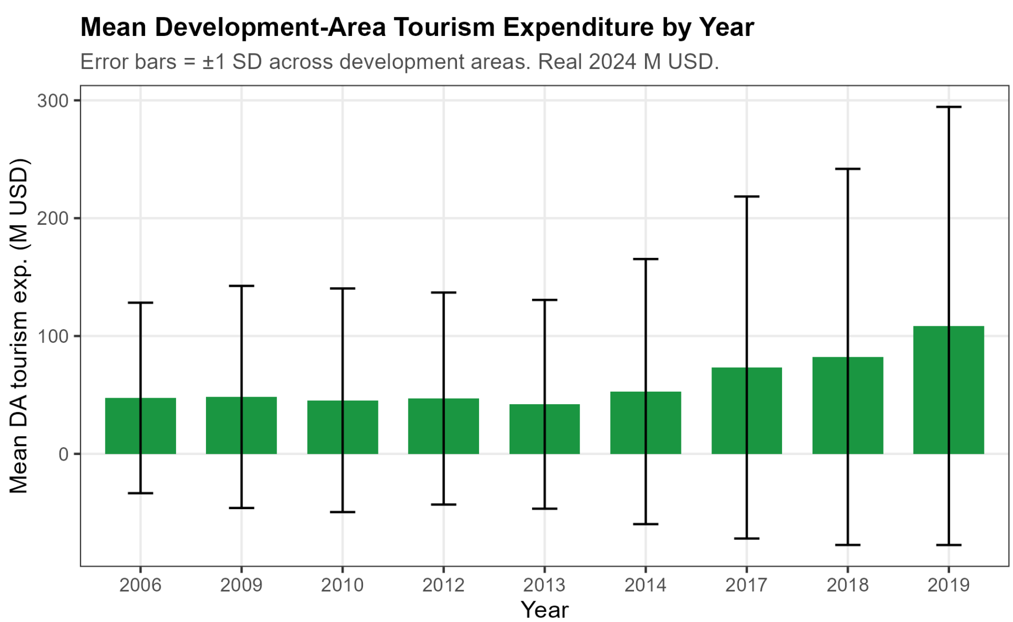

Figure 1: Mean Development-Area Tourism Expenditure by Year
Mean real accommodation expenditure (2024 USD) per development area by JSLC survey year. Error bars show ±1 standard deviation across development areas.

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2.2 The Jamaican Education System

Schooling in Jamaica is compulsory from primary school beginning in grade 1 up until grade 11.[5] The system is divided into primary and secondary stages of schooling, with primary schools teaching grades 1 through 6, and secondary schools teaching grades 7 through 13.[6] The crucial milestone in the secondary stage that helps determine college attendance is the Caribbean Secondary Education Certificate examination, taken in the 11th grade.[7] These exams cover five major subject areas and are the primary secondary credential.[8] These examinations provide entry to skilled work, and are a necessary pre-requisite for those who want to continue on to university.

Successful completion of the CSEC exams grants access to Sixth Form: two years of additional study in three or four subject areas designed to prepare students for university. In the U.S. context, this is roughly equivalent to Advanced Placement or International Baccalaureate coursework. At the end of Sixth Form (grade 13), students sit the Caribbean Advanced Proficiency Exam (CAPE).

Primary and secondary schooling is largely administered by the government and nominally free, while early childhood education is primarily provided by private institutions with some government support. Significant costs remain for families, particularly at the secondary level, including uniforms, textbooks, transportation, and meals. Households spend the equivalent of 3 percent of GDP on early childhood, primary, and secondary education. Relative to the average GDP per capita in 2017 of approximately 3,053 USD, the household cost for one child represented 13 percent of GDP per capita for primary school and 20 percent for secondary education.[9] Financial constraints are accordingly the most commonly cited reason for leaving school before grade 11, as shown in Table **??**, with lunch, snacks, and transportation comprising the largest reported shares of schooling expenditures.

[5] Enrollment in early childhood education is not required.

[6] Source: (*Public Expenditure Review of the Education Sector in Jamaica* 2021)

[7] This is based upon the British O-Level Examination. Completion of these exams is equivalent to receiving a high school diploma in the United States.

[8] Source: CSEC Equivalent in USA – American Education & Translation Services (AET)

[9] Source:

The government invests considerably in the education system, including through cash transfer programs to encourage attendance[10], Ministry of Education-funded lunch and snack programs[11], and support for low-income families to cover textbooks and other materials.[12] Jamaica has spent more than many peer nations on education, with the education budget reaching approximately 5 percent of GDP. These supports may reduce dropout rates by easing the financial constraints facing families.

There is near-universal school attendance up to age 16 (World Bank 2022), but Jamaican students perform worse than those in peer nations on measures of learning.[13] Jamaican students complete an average of 11.4 years of formal education, but only 7.1 Learning-Adjusted Years of Schooling (LAYS) (World Bank 2022), reflecting significant gaps in schooling quality. Another key challenge is secondary school completion and tertiary enrollment: approximately 90 percent of students complete secondary education, while only 27 percent enroll in tertiary institutions. The transition from secondary to tertiary education is therefore one of the most important margins to examine.

2.3 Hypotheses Regarding Tourism and Human Capital Investments

The literature on educational attainment stresses the importance of contemporaneous schooling costs for household and individual decisions about optimal education levels. In the Jamaican context, the main predictions are that, in the presence of significant household financial constraints, a positive shock to local employment and earnings will influence schooling by raising consumption, improving future human capital, decreasing the marginal utility of additional consumption, and potentially altering the net returns to schooling. I derive the following hypotheses:

1. Decreased enrollment and attendance, and an increase in students dropping out to join the labor force.

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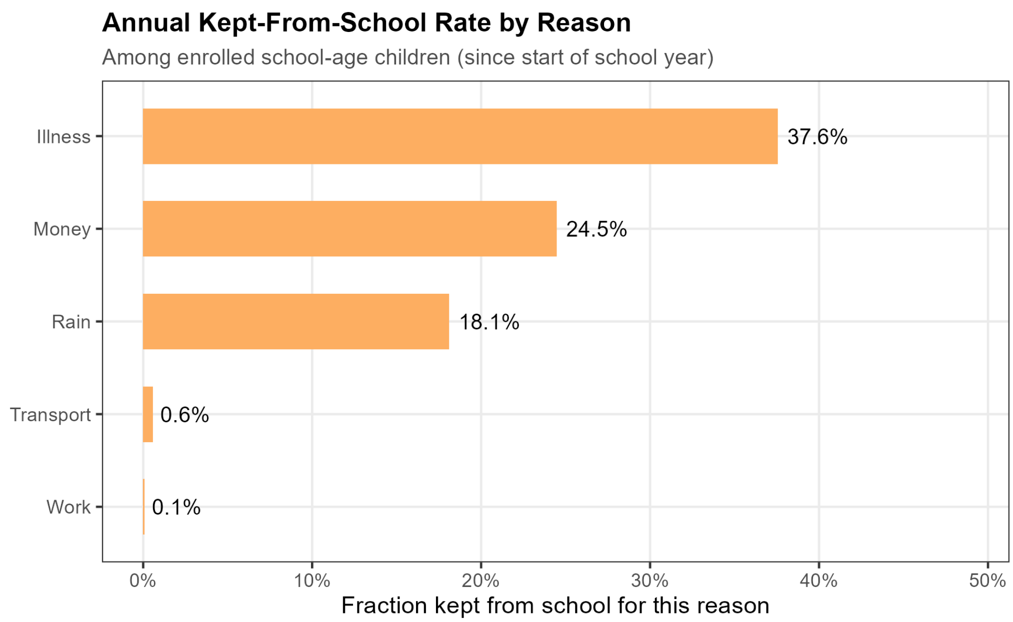

Figure 2: Annual Kept-From-School Rate by Reason
Share of enrolled school-age children kept from school for each reason at least once since the start of the school year, pooled across JSLC waves.

2. For students who remain in school, a greater share sorting into vocational or skills training programs and fewer choosing to pursue a university degree.

The first hypothesis reflects the finding in McKetty (2026) that increases in real expenditures resulting from higher tourism levels are approximately 1.8 percent per year and are heavily concentrated in food and healthcare. The utility gained from greater consumption and improved health may be perceived as more valuable than additional investment in schooling, particularly given the high investment threshold required to meaningfully improve educational performance. The second hypothesis reflects the expectation that, because of greater opportunities in vocational and technical fields and the high cost of college, more students will sort into those tracks rather than pursuing formal higher education.

The first hypothesis stems from the fact that from the paper McKetty (2026) I know that the increase in real expenditures resulting from higher tourism levels is roughly 1.8 percent per year and is heavily spent on food and healthcare. The utility gained from consuming more and improving health for the future may be seen as more useful than additional investments in schooling given a potentially high threshold of investment needed to improve performance.

Essentially, households will believe that too much investment is needed to make school a worthy route, and the increased earnings opportunities make work the better option. For hypothesis 2, I expect that because of greater opportunities in vocational or technical skills training, and because of the exorbitant cost of college education, more students will sort into those tracks than before and less towards college studies.

3 Data

Data on tourist expenditures come from exit surveys conducted by the Ministry of Tourism on a sample of departing tourists leaving through one of the island's two international airports. Enumerators randomly select traveling parties to survey in the departure halls. Because all stopover tourists (visitors who spend more than 24 hours in Jamaica) must depart through these airports, the surveys obtain a representative sample of non-cruise visitors. Respondents answer detailed questions about their spending, demographics, and, critically, the hotel at which they stayed. This provides a representative picture of the distribution of tourist spending across Jamaican locales. I scale the survey estimates using aggregate arrival and spending data from annual reports published by the Ministry of Tourism. Tourism data span the years 2004–2006 and 2008–2019, covering approximately 50,000 traveling parties and 100,000 individuals.

Education data come from the Jamaica Survey of Living Conditions, a nationally representative household survey conducted annually by the Statistical Institute of Jamaica. The JSLC collects questions at both the household and individual levels. In the education module, questions on enrollment are asked for all individuals. For currently enrolled children and youth, detailed questions cover the type of school attended, academic progress and exam performance, and spending by parents across categories such as tuition, textbooks, and transportation. Individuals no longer in school answer questions about their educational attainment, skill levels, and, for those who dropped out before grade 11, their reasons for doing so. The JSLC years used in this paper are 2006, 2010, 2012, 2014, 2018, and 2019, covering approximately 6,000 households and 11,000 school-aged children. Administrative shapefiles from STATIN enable me to link tourism spending and households across spatially consistent geographic units.

4 Empirical Approach

I calculate tourism at the development-area level because development areas are government-designated districts with similar economic and demographic characteristics. These areas provide a natural grouping for communities and economic activities, and include both rural and urban components, allowing for estimation of heterogeneous rural-urban impacts. The main empirical specification is the following, where is an education outcome variable, is a vector of household controls, and is the lagged level of tourism. I cluster standard errors at the development-area level, and include development-area and year fixed effects (and , respectively). I use a one-year lag of tourism revenue, reflecting the empirical finding in the literature that schooling decisions are influenced by conditions in prior periods. I do not include additional lags because of likely serial correlation between immediately adjacent years. I run robustness checks using contemporaneous and two-year lags.

4.1 The Shift-Share Instrumental Variable Approach

The central empirical challenge is that the level of tourism activity in a development area may be correlated with local economic conditions that independently affect household education decisions. To address this endogeneity, I employ a shift-share (Bartik) instrumental variable strategy.

The instrument is constructed by interacting aggregate "shifts" — year-to-year changes in the number of tourists from each origin country o arriving in Jamaica, ΔN_{ot} — with predetermined "shares" capturing each development area d's historical exposure to tourists from origin country o, s_{od}. The Bartik instrument is:

$$B_{dt} = \sum_{o} s_{od} \cdot \Delta N_{ot} \tag{1}$$

where s_{od} is the share of development area d's tourist accommodation expenditures attributable to origin country o in a base year, and ΔN_{ot} is the national-level change in arrivals from origin country o in year t. The identifying assumption is that the national-level shifts in origin-country arrivals are orthogonal to local development-area shocks to education outcomes, conditional on development area and year fixed effects.

The baseline estimating equation is:

$$Y_{idt} = \alpha + \beta \, \widehat{\text{Tourism}}_{dt} + \mathbf{X}'_{idt}\gamma + \delta_d + \lambda_t + \varepsilon_{idt} \tag{2}$$

where Y_{idt} is an education outcome for individual i in development area d in year t; $\widehat{\text{Tourism}}_{dt}$ is the fitted value of local tourism accommodation expenditures from the first stage using B_{dt} as the excluded instrument; $\mathbf{X}_{idt}$ is a vector of individual and household controls; δ_d and λ_t are development area and year fixed effects; and ε_{idt} is a disturbance term clustered at the development area level.

I use a one-year lag of tourism revenue, reflecting the empirical finding in the literature that schooling decisions are influenced by conditions in prior periods. I do not include additional lags because of likely serial correlation between immediately adjacent years. I run robustness checks using contemporaneous and two-year lags.

Development-area tourism levels are likely correlated with unobserved features of these locales that also influence educational outcomes. More touristic development areas may, for example, have systematically better governance, resulting in both higher-quality schools and greater hotel investment. Such endogeneity would bias OLS estimates. To address this, I instrument area revenues using a shift-share instrumental variable. I exploit year-over-year variation in the intensity of tourist arrivals from different regions and countries, interacted with the heterogeneous preferences of tourists from different origin countries for different parts of Jamaica.

The shift-share IV is constructed by interacting an exposure "share" component with aggregate "shifts" for each development area. The exposure share for development area d is the share of its tourism revenues in a base year t_0 earned from tourists originating from region r. The shift measures the percentage change in spending by tourists from region r across all of Jamaica (excluding area d) between year $t-1$ and the base year t_0. Exogeneity can be established either by arguing that the base-year exposure shares are orthogonal to the outcome or that the shocks themselves are orthogonal. This paper uses the shifts-based identification approach, which is the most appropriate given the context of tourist decision-making. Decisions by tourists to visit Jamaica are influenced by home-country economic conditions, conditions in competing destinations, and other factors that, conditional on year and area fixed effects, are plausibly exogenous to the development-area characteristics that influence educational investment.

I estimate the effect of tourism on schooling using two-stage least squares. Following recommendations in the literature, I residualize the shocks to correct for serial correlation. To address potential correlation in shocks across development areas with similar exposure shares, I conduct robustness checks using Adão et al. (2019) shift-level regressions.

5 Results

I first present findings on the effects of one-year lagged tourism levels on enrollment, attendance, and reasons for dropping out. I then discuss results for schooling expenditures. Finally, I present findings on educational attainment and exam performance.

5.1 Baseline Findings: Enrollment and Dropping Out

I first consider the effects of lagged local tourism on the probability that students are enrolled at three stages of schooling. The results are provided in Table 1. I find no effect of prior-year development-area tourism expenditures on the likelihood that a child or young adult is enrolled in grade school at any level. Column 2 then limits the sample to those between the ages of 14 and 19, and again finds no effect of previous-year tourism expenditures on whether or not a teenager is enrolled in college preparatory 6th form coursework. All regressions include year and development-area fixed effects. In the final column I consider the role of tourism on college enrollment, and find that lagged tourism earnings result in a lower likelihood that students are enrolled in college. This suggests that booms in tourism act at the transition out of secondary school and into higher education.

Table 2 presents regressions examining the effect of lagged tourism levels on the reasons non-enrolled students are not attending school, and finds no effect for any of the outcomes examined: money problems, family reasons, lack of interest, and pregnancy.

Table 1: School Enrollment, IV Estimates

Dependent Variables:	Enrolled in school	Currently in sixth form	Currently in college
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.007	0.034	-0.065**
lag 1 (100M USD)	(0.036)	(0.034)	(0.031)
Female	0.006*	0.007	0.045***
	(0.003)	(0.006)	(0.007)
Age	-0.038***	0.004*	-0.006***
	(0.001)	(0.002)	(0.001)
No. enrolled siblings (HH)	-0.0002	-0.002	-0.006**
	(0.002)	(0.003)	(0.003)
Fixed-effects			
Development area	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Bartik F-stat (1st stage)	1,030.2	446.07	567.73
Observations	13,518	5,494	5,985
R^2	0.25524	0.16714	0.03906
Sample			
School-age (5–19)	Yes	No	No
Upper secondary (14–19)	No	Yes	No
College-age (17–27)	No	No	Yes
Mean of dep. var.	0.879	0.062	0.070

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 3 covers the effects of lagged area tourism revenues on the frequency and reasons for absences. Column 1 shows that prior-year tourism revenues produce a statistically significant decrease in the likelihood that an enrolled student is absent occasionally or frequently for any reason: a 1 million dollar increase produces a 0.2 percent decrease in the likelihood of frequent or occasional absence. Column 4 indicates that this effect is driven by a decrease in illness-related absences, suggesting that additional income for local households is enabling better access to medical services. This is consistent with results from McKetty (2026), which showed that higher local tourism revenues resulted in a real increase in household spending on healthcare.

Table 2: Dropout Reasons, IV Estimates

Dependent Variables:	Dropout: financial	Dropout: family	Dropout: no interest	Dropout: pregnancy
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp., lag 1 (100M USD)	0.030	0.023	-0.016	0.030
	(0.042)	(0.027)	(0.050)	(0.052)
Fit statistics				
Bartik F-stat (1st stage)	172.38	172.38	172.38	172.38
Observations	1,637	1,637	1,637	1,637
R^2	0.08484	0.05091	0.04512	0.07579
Mean of dep. var.	0.038	0.020	0.043	0.044

Notes: Sample: children ages 5–19 not currently enrolled. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Interestingly, the table also shows that higher tourism is associated with a small but precisely estimated increase in the likelihood that students miss school occasionally or frequently because of transportation costs. Table 9 shows that this effect is entirely driven by rural students. This may reflect localized inflationary pressure on transportation services resulting from area tourism activity, effectively "crowding out" local students.[14] While the effect of higher local tourism on illness-related absenteeism outweighs the increase in transportation-related absences, this finding highlights an important potential spillover of greater specialization in local tourism services.

Work by Allen et al. (2021) showed how tourist spending in specific neighborhoods of

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Table 3: School Absence Reasons, IV Estimates

Dependent Variables:	Kept out, any reason	Kept out, money (ever)	Kept out, transport	Kept out, illness	Kept out, work
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (100M USD)	-0.176**	0.075	0.020***	-0.165**	0.002
	(0.086)	(0.109)	(0.006)	(0.079)	(0.002)
Fit statistics					
Bartik F-stat (1st stage)	863.27	863.27	863.27	863.27	863.27
Observations	11,881	11,881	11,881	11,881	11,881
R^2	0.11458	0.05459	0.00211	0.02904	0.00628
Mean of dep. var.	0.188	0.259	0.003	0.063	0.001

Notes: Sample: enrolled children; absence measured by annual recall (occasional or frequent). Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Barcelona crowded out local consumption in those areas. The finding here points to an analogous effect operating through local transportation services. This channel is particularly relevant given the spatial inequalities between urban and rural communities in Jamaica, where rural households are considerably poorer[15] and must travel further to access education. Tourism growth, while boosting incomes for urban households, may also exacerbate inequality by pricing out those who are most reliant on transportation services and most likely to be poor and geographically distant from schools.

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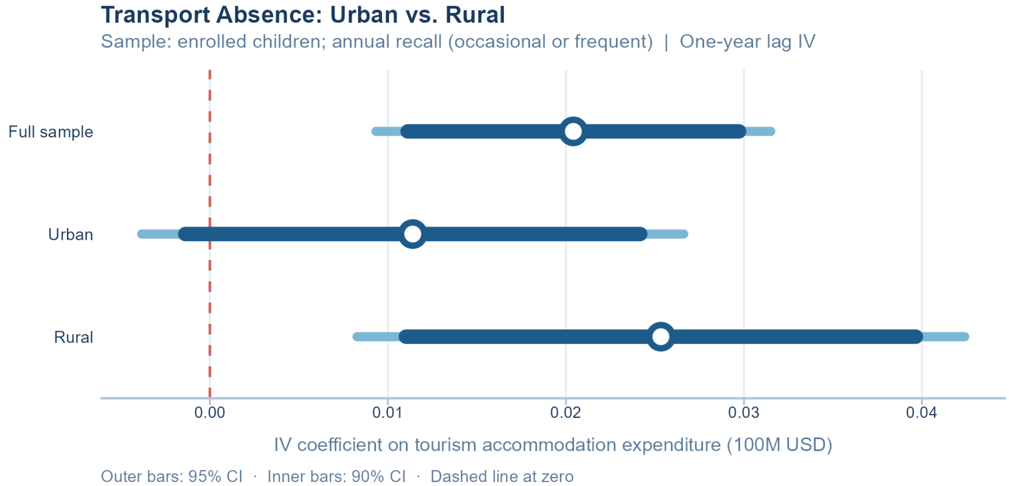

Figure 3: Transport Absence, Urban vs. Rural: IV Coefficient Estimates
Tourism expenditure coefficients (one-year lag) from the IV regressions in Table 9. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

5.2 School Expenditures

Table 4: School Expenditure by Category, IV Estimates

Dependent Variables:	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 1 (100M USD)	-20.6	-24.5**	59.8	-53.5	48.8
	(33.9)	(9.83)	(65.5)	(32.9)	(79.2)
Fit statistics					
Bartik F-stat (1st stage)	431.64	634.72	638.01	657.07	632.38
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.04607	0.08583	0.07780	0.03300	0.12224
Mean of dep. var.	48.388	70.385	258.767	88.556	464.417

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 5: Examination Attainment, IV Estimates

Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level	Degree
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp., lag 1 (100M USD)	-0.007 (0.016)	-0.019 (0.012)	-0.007 (0.013)	0.005 (0.015)	-0.034** (0.014)	-0.004 (0.005)
Fit statistics						
Bartik F-stat (1st stage)	1,030.2	1,030.2	1,030.2	1,030.2	484.43	484.43
Observations	13,518	13,518	13,518	13,518	6,019	6,019
R^2	0.14591	0.13809	0.11882	0.05256	0.02929	0.00777
Sample						
School-age (5–19)	Yes	Yes	Yes	Yes	No	No
Secondary/College (14–20)	No	No	No	No	Yes	Yes
Mean of dep. var.	0.061	0.056	0.046	0.020	0.017	0.001

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 16–25 for A-Level and Degree. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

5.3 Educational Attainment and Exam Performance

The final set of results covers the effects of area tourism levels on the educational tracks that individuals pursue. Table 5 regresses indicators for completing different exams on prior-year development-area tourism levels. The exams examined are, in order, the Grade Nine Achievement Test (GNAT), the CXC Basic Exam, the CXC General, and an indicator for having passed both the math and English components.[16] There is no effect across any of these examination levels, indicating that lagged tourism earnings do not affect the sorting of students across academic tracks at the high school level.

[16] The GNAT was used to place students into secondary high schools.

6 Discussion and Conclusion

This paper has studied the relationship between tourism-sector specialization and educational attainment and investment by households in Jamaica. I combined granular surveys of tourist spending with household survey data covering absenteeism, school supply expenditures, and academic performance from primary through tertiary levels of education. Using regression analysis, I evaluate the effect of lagged area-level tourism earnings on enrollment, attendance, selection into different academic tracks, and financial investments in schooling.

I find that lagged tourism largely produces little to no effect on schooling attainment and household investment across academic stages. I estimate precise zeros for the effects of tourism on enrollment across grade-school levels. Prior-year tourism levels also have little effect on the reasons given for not being in school, with a notably absent effect on the likelihood of dropping out because of financial problems. I do, however, observe some statistically significant effects on absenteeism: higher prior-year tourism levels are associated with a 0.2 percent decrease in frequent or occasional absences, driven by a reduction in illness-related absences. This is consistent with findings from McKetty (2026), which showed that tourism resulted in increased household healthcare spending. On the other hand, greater development-area tourist activity is associated with a 0.02 percent increase in the likelihood of occasional or more frequent absences due to transportation costs, an effect concentrated entirely among rural households. Additionally, greater tourism expenditures are associated with lower textbook spending and a lower likelihood of college enrollment.

The analysis also reveals ways in which higher tourism activity can reduce educational investment and attainment. Higher lagged tourism levels are associated with lower textbook spending, driven by a statistically significant increase in the share of enrolled students who spend nothing on textbooks. One interpretation is that this reflects a lower perceived need for educational materials given reduced incentives to pursue formal education. This interpretation is somewhat tempered by the absence of any effect of tourism on sixth-form enrollment. The negative effect on college enrollment may indicate that tourism opportunities reduce attainment primarily at the margin between tertiary education and entering the workforce.

The finding that tourism is associated with a small but statistically significant increase in transportation-related absences among rural students points to an important mechanism by which tourism specialization—and service-sector specialization more broadly, given the in-person nature of service consumption—can exacerbate existing inequalities. The crowding out of local transportation by tourism is consistent with prior work on the sector and provides evidence of a channel through which positive shocks to local sectors can negatively affect educational attainment.

The core takeaway is that tourism-sector specialization has primarily little to no effect on educational attainment and investment in Jamaica, and that the observed effects are mixed. There is no evidence that lagged tourism levels increase the likelihood of students dropping out during grade school. Through better access to healthcare, some children may improve attendance, which is important for academic achievement; but potential inflationary pressure on transportation may harm attendance for rural students. While dropping out appears largely unaffected, there may be an effect at the margin between secondary schooling and tertiary education.

These findings depart from those of prior work on development and human capital accumulation. Unlike studies such as Shah and Steinberg (2017) and Atkin (2016), there is no evidence here that greater tourism activity leads more students to leave school. Reinforcing this, I observe no effect of lagged tourism on students missing school to work, and no change in the probability of leaving school because of financial constraints. Instead, the analysis reveals that growth in a sector dominated by middle- and lower-skilled labor can marginally improve attendance through better access to healthcare.[17] This pathway differs from the more commonly observed income-channel effects, in which higher household earnings reduce the need for children to work or free up additional resources for schooling. Here, more income enables better medical care, which in turn reduces illness-related absences.

Another distinguishing feature of the results is the concentration of potential substitution-channel effects at the margin between sixth form and college. The negative effect of lagged tourism on college enrollment may be consistent with substitution from higher education to

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the workforce. This interpretation is reinforced by the absence of any effect of tourism on enrollment in skills or training programs. The negative effect on textbook spending could also reflect a lowering of investment in materials that support continued study. Firmer conclusions are difficult to draw from the estimated effects alone.

This study cannot directly explain the absence of broader effects on academic outcomes, but several factors are worth considering. First, unemployment was persistently high during the study period, averaging 10 percent between 2010 and 2019. There may therefore have been limited work available for children and young adults in the tourism sector's mid- and lower-skilled jobs given the large supply of excess labor. Second, the quality of tourism-sector jobs may be accurately perceived as low in terms of wages and advancement potential.[18] Discontentment among tourism workers has been well documented in the Jamaican media. Third, because tourism is largely composed of mid-skilled jobs requiring completion of 11th grade, a boom in tourism services may not reduce the likelihood of completing secondary education, but instead affect whether students choose to pursue tertiary education. The real challenge may be that gains in mid-skilled services leave secondary school completion largely unchanged—or even strengthen it—while limiting incentives to progress further, producing a kind of "secondary education trap" that could prove challenging for the long-term growth of middle-income nations like Jamaica.

The growth of human capital in middle-income countries depends heavily on micro-level household decisions about how much education to obtain and how much to invest at each stage. Tourism services are of growing importance in these economies, but we know little about how the growth of this industry alters household incentives regarding education at both the extensive and intensive margins. The results of this paper demonstrate the potential for income effects to operate through channels such as health, and for substitution effects to operate at the margin between secondary schooling and university education. Specific to tourism and other consumer services, the data also show how local sector booms can alter the provision of services in local communities, potentially disadvantaging the most vulnerable.

For policymakers considering tourism specialization as a development strategy, the results

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on local service provision underscore the need to consider how booms in sectors that require in-person consumption can produce positive spillovers for some while creating challenges for others. For researchers, a productive line of inquiry would examine the distributional impacts of different service-sector categories on human capital accumulation, with attention to how crowding out of local services may constrain educational access through effects on transportation costs or other goods and services. Such work could illuminate a tension between the income effects of local service-sector expansion and localized inflation.

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7 Appendix

7.1 Additional Main Specification Results

Table 6: Skills Program Enrollment and Completion, IV Estimates

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp., lag 1 (100M USD)	-0.006	-0.006	0.040
	(0.010)	(0.010)	(0.039)
Fit statistics			
Bartik F-stat (1st stage)	114.72	114.72	189.77
Observations	1,576	1,576	1,866
R^2	0.14540	0.14540	0.14185
Mean of dep. var.	0.003	0.003	0.077

Notes: Sample: children ages 5–19. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

7.2 Instrument Summary Statistics and First Stage Regressions

[Appendix placeholder.]

7.3 Shift-Share Instrument Diagnostics and Summary Statistics

[Appendix placeholder.]

7.4 Additional Regression Results: Contemporaneous Specification

Tables 10–16 replicate the main results using a contemporaneous specification in which development area tourism expenditures and the Bartik instrument enter without a lag.

Table 7: School Attendance, IV Estimates

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.245	-0.245	0.043
lag 1 (100M USD)	(0.721)	(0.721)	(0.036)
Fit statistics			
Bartik F-stat (1st stage)	862.81	862.81	863.27
Observations	11,412	11,412	11,881
R^2	0.03435	0.03435	0.22997
Mean of dep. var.	18.803	18.803	0.139

Notes: Sample: currently enrolled children ages 5–19. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 8: School Expenditure by Category (Log), IV Estimates

Dependent Variables:	Log tuition	Log books	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	-0.596	-4.01**	2.29	-1.27	1.32
lag 1 (100M USD)	(1.95)	(1.98)	(2.18)	(0.960)	(1.05)
Fit statistics					
Bartik F-stat (1st stage)	431.64	634.72	638.01	657.07	632.38
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.08843	0.06796	0.13568	0.02279	0.02790
Mean of dep. var.	-17.676	-0.619	-2.766	2.586	4.455

Notes: Sample: enrolled children; 2014 excluded due to survey design. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

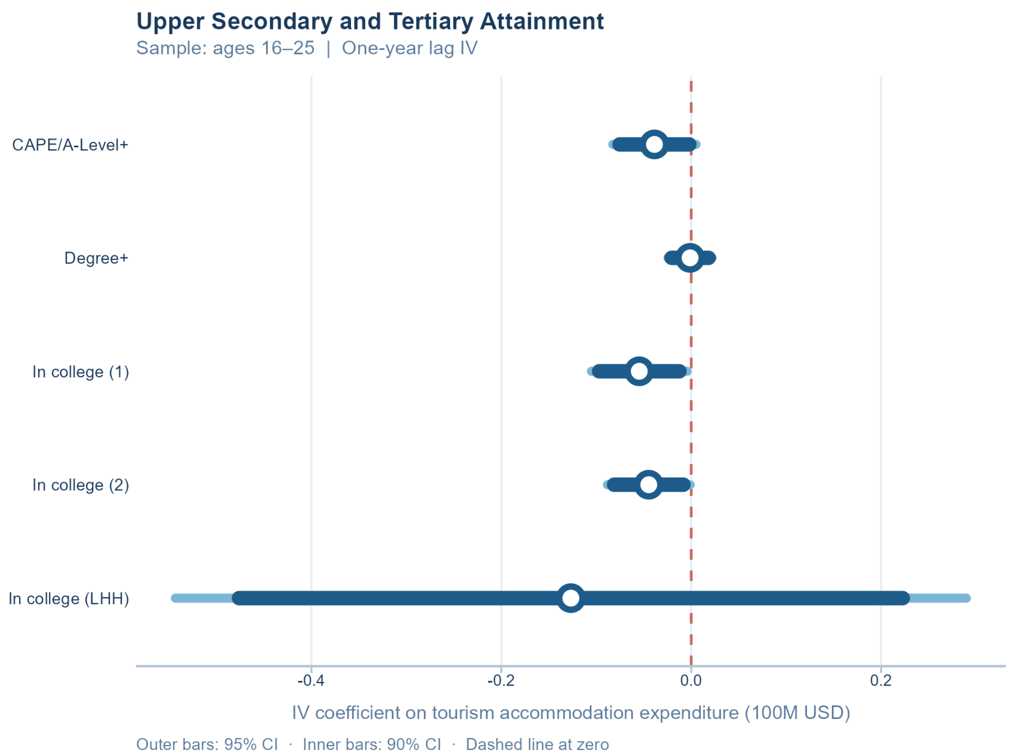

Figure 4: Upper Secondary and Tertiary Attainment: IV Coefficient Estimates
Tourism expenditure coefficients (one-year lag) from the IV regressions in Table **??**. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

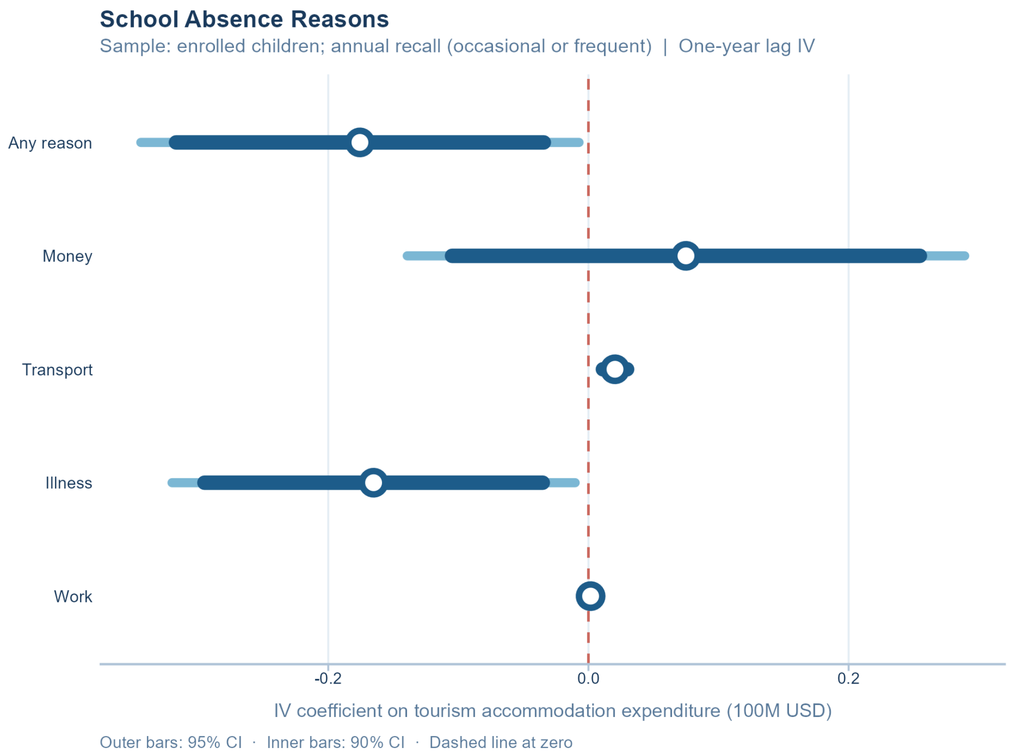

Figure 5: Absence Reasons: IV Coefficient Estimates

Tourism expenditure coefficients (one-year lag) from the IV regressions in Table 3. Wide bars show 95% confidence intervals; narrow bars show 90% intervals. Dashed line marks zero.

Table 9: Transport Absence by Urban/Rural, IV Estimates

Dependent Variable:		Kept out, transport	
Model:	Full Sample	Urban HHs	Rural HHs
Variables			
Tourism exp.,	0.020***	0.011	0.025***
lag 1 (100M USD)	(0.006)	(0.008)	(0.009)
Fit statistics			
Bartik F-stat (1st stage)	863.27	190.86	2,864.2
Observations	11,881	4,501	7,380
R^2	0.00211	-0.00407	0.02772
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. Controls for gender, age, and enrolled household siblings are included in all specifications but not shown. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 10: School Enrollment, IV Estimates (Contemporaneous)

Dependent Variable:	Enrolled in school	
Model:	(1)	(2)
Variables		
Tourism exp.	-0.247	-0.247
(100M USD)	(0.404)	(0.404)
Female	0.006*	0.006*
	(0.003)	(0.003)
Age	-0.038***	-0.038***
	(0.0010)	(0.0010)
No. enrolled siblings (HH)	0.002	0.002
	(0.004)	(0.004)
Fit statistics		
Bartik F-stat (1st stage)	34.270	34.270
Observations	13,687	13,687
R^2	0.17776	0.17776
Mean of dep. var.	0.879	0.879

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 11: Dropout Reasons, IV Estimates (Contemporaneous)

Dependent Variables:	Dropout: financial	Dropout: family	Dropout: no interest	Dropout: pregnancy
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.	-0.386	-0.070	-0.004	0.315
(100M USD)	(0.361)	(0.154)	(0.184)	(0.310)
Female	-0.034***	-0.005	-0.041***	0.081***
	(0.013)	(0.009)	(0.009)	(0.011)
Age	-0.002	-0.004**	-0.005	-0.004*
	(0.003)	(0.002)	(0.003)	(0.002)
No. enrolled siblings (HH)	0.006	-0.004	-0.005	0.011*
	(0.006)	(0.003)	(0.003)	(0.006)
Fit statistics				
Bartik F-stat (1st stage)	5.6526	5.6526	5.6526	5.6526
Observations	1,655	1,655	1,655	1,655
R^2	-0.37923	0.00446	0.04463	-0.20542
Mean of dep. var.	0.039	0.020	0.044	0.044

Notes: Sample: children ages 5–19 not currently enrolled. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 12: School Attendance, IV Estimates (Contemporaneous)

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	1.03	1.03	0.097
(100M USD)	(4.16)	(4.16)	(0.365)
Female	0.052	0.052	0.001
	(0.113)	(0.113)	(0.005)
Age	-0.026**	-0.026**	-0.007***
	(0.011)	(0.011)	(0.0010)
No. enrolled siblings (HH)	-0.044	-0.044	-0.007
	(0.079)	(0.079)	(0.006)
Fit statistics			
Bartik F-stat (1st stage)	27.989	27.989	29.229
Observations	11,557	11,557	12,032
R^2	0.02671	0.02671	0.22087
Mean of dep. var.	18.804	18.804	0.138

Notes: Sample: currently enrolled children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 13: School Absence Reasons, IV Estimates (Contemporaneous)

Dependent Variables:	Kept out, any reason	Kept out, money	Kept out, transport	Kept out, illness	Kept out, work
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-0.314	-0.124	0.085	-0.298	-0.004
(100M USD)	(0.371)	(0.352)	(0.088)	(0.247)	(0.014)
Female	-0.027***	-0.018***	-0.0001	-0.002	-0.0004
	(0.007)	(0.007)	(0.001)	(0.004)	(0.0004)
Age	-0.002	0.002	0.0005***	-0.004***	3.07×10^{-5}
	(0.001)	(0.001)	(0.0002)	(0.0005)	(5.68×10^{-5})
No. enrolled siblings (HH)	0.019***	0.023***	-0.0004	0.0003	-5.11×10^{-5}
	(0.006)	(0.006)	(0.0010)	(0.003)	(0.0002)
Fit statistics					
Bartik F-stat (1st stage)	29.229	29.229	29.229	29.229	29.229
Observations	12,032	12,032	12,032	12,032	12,032
R^2	0.04245	0.04741	-0.31279	-0.14018	0.00062
Mean of dep. var.	0.190	0.108	0.003	0.063	0.000

Notes: Sample: enrolled children; absence measured by annual recall (occasional or frequent). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 14: Transport Absence by Urban/Rural, IV Estimates (Contemporaneous)

Dependent Variable:		Kept out, transport	
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	0.085	0.023	0.224
(100M USD)	(0.088)	(0.023)	(0.387)
Female	-0.0001	-0.002	-0.0001
	(0.001)	(0.002)	(0.002)
Age	0.0005***	0.0005**	0.0004*
	(0.0002)	(0.0002)	(0.0002)
No. enrolled siblings (HH)	-0.0004	0.0010	-0.002
	(0.0010)	(0.001)	(0.003)
Fit statistics			
Bartik F-stat (1st stage)	29.229	22.257	15.899
Observations	12,032	4,501	7,531
R^2	-0.31279	-0.06044	-0.48178
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 15: School Expenditure by Category, IV Estimates (Contemporaneous)

Dependent Variables:	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-72.1	-157.4	52.6	-133.7	-71.1
(100M USD)	(113.5)	(181.5)	(146.5)	(142.7)	(235.3)
Female	-3.30	6.32***	24.9***	-7.42***	15.5**
	(6.17)	(1.75)	(7.92)	(1.78)	(6.82)
Age	-7.25***	1.03***	31.5***	2.61***	29.1***
	(0.800)	(0.247)	(2.51)	(0.265)	(1.24)
No. enrolled siblings (HH)	-11.7***	-7.50***	-12.3***	-6.69***	-26.7***
	(2.69)	(1.99)	(3.14)	(1.49)	(3.99)
Fit statistics					
Bartik F-stat (1st stage)	28.389	18.818	21.921	21.131	19.856
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.03084	-0.54049	0.07799	-0.18035	0.12014
Mean of dep. var.	47.787	70.273	257.486	88.431	463.130

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 16: School Expenditure by Category (Log), IV Estimates (Contemporaneous)

Dependent Variables:	Log tuition	Log books	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.	-5.52	-8.51	6.60	-0.261	-0.244
(100M USD)	(12.0)	(11.6)	(7.23)	(2.90)	(3.74)
Female	0.049	0.695***	0.901***	0.085	-0.070
	(0.194)	(0.193)	(0.231)	(0.138)	(0.111)
Age	-0.550***	-0.334***	1.11***	0.020	0.179***
	(0.034)	(0.037)	(0.041)	(0.020)	(0.019)
No. enrolled siblings (HH)	-0.324*	-0.796***	-0.438***	-0.232**	-0.192***
	(0.162)	(0.130)	(0.129)	(0.102)	(0.060)
Fit statistics					
Bartik F-stat (1st stage)	28.389	18.818	21.921	21.131	19.856
Observations	9,384	10,427	10,429	10,448	10,483
R^2	0.01559	-0.02140	0.08859	0.02339	0.03028
Mean of dep. var.	-17.707	-0.628	-2.791	2.584	4.454

Notes: Sample: enrolled children; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 17: Examination Attainment, IV Estimates (Contemporaneous)

Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level	Degree
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp.	0.097	0.082	0.051	0.005	-0.234	-0.025
(100M USD)	(0.174)	(0.155)	(0.110)	(0.066)	(0.204)	(0.049)
Female	0.032***	0.031***	0.026***	0.011***	0.015***	0.004
	(0.004)	(0.004)	(0.004)	(0.002)	(0.006)	(0.003)
Age	0.021***	0.019***	0.016***	0.007***	0.010***	0.003***
	(0.0008)	(0.0009)	(0.0009)	(0.0008)	(0.001)	(0.0007)
No. enrolled siblings (HH)	-0.006**	-0.006**	-0.006**	-0.002**	-0.006**	-0.001
	(0.003)	(0.003)	(0.003)	(0.0009)	(0.002)	(0.0009)
Fit statistics						
Bartik F-stat (1st stage)	34.270	34.270	34.270	34.270	23.957	23.957
Observations	13,687	13,687	13,687	13,687	6,275	6,275
R^2	0.11670	0.11635	0.10554	0.05221	-0.14401	0.01262
Sample						
School-age (5–19)	Yes	Yes	Yes	Yes	No	No
Upper secondary (16–25)	No	No	No	No	Yes	Yes
Mean of dep. var.	0.061	0.056	0.046	0.019	0.041	0.006

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 16–25 for A-Level and Degree. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 18: Upper Secondary and Tertiary Attainment, IV Estimates (Contemporaneous)

Dependent Variables:	At least CAPE/A-Level	At least degree	Currently in college		
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp. (100M USD)	-0.182	0.008	-0.092	-0.123	0.108
	(0.178)	(0.022)	(0.107)	(0.142)	(0.227)
Female	0.029***	0.009***	0.036***	0.036***	0.032*
	(0.005)	(0.002)	(0.006)	(0.007)	(0.017)
Age	0.010***	0.004***	-0.0001	0.001	-0.0005
	(0.002)	(0.0008)	(0.0008)	(0.001)	(0.002)
No. enrolled siblings (HH)				-0.005*	
				(0.003)	
Fit statistics					
Bartik F-stat (1st stage)	32.058	32.058	32.058	23.957	5.8504
Observations	7,879	7,879	7,879	6,275	315
R^2	-0.04551	0.02229	0.01975	0.01054	0.26013
Mean of dep. var.	0.049	0.010	0.067	0.065	0.029

Notes: Sample: ages 16–25. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 19: Skills Program Enrollment and Completion, IV Estimates (Contemporaneous)

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp.	-0.013	-0.013	0.125
(100M USD)	(0.041)	(0.041)	(0.174)
Female	0.003	0.003	-0.003
	(0.002)	(0.002)	(0.011)
Age	-0.0006	-0.0006	0.014***
	(0.0006)	(0.0006)	(0.004)
No. enrolled siblings (HH)	-0.001*	-0.001*	-0.009*
	(0.0006)	(0.0006)	(0.005)
Fit statistics			
Bartik F-stat (1st stage)	3.5134	3.5134	5.6541
Observations	1,594	1,594	1,886
R^2	0.13597	0.13597	0.12238
Mean of dep. var.	0.003	0.003	0.076

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

7.5 Additional Regression Results: Two-Year Lagged Specification

Tables 20–26 replicate the main results using a two-year lag specification in which development area tourism expenditures and the Bartik instrument enter with a two-year lag.

Table 20: School Enrollment, IV Estimates (Lag 2)

Dependent Variable:	Enrolled in school	
Model:	(1)	(2)
Variables		
Tourism exp.,	-0.044	-0.044
lag 2 (100M USD)	(0.071)	(0.071)
Female	0.006*	0.006*
	(0.003)	(0.003)
Age	-0.038***	-0.038***
	(0.001)	(0.001)
No. enrolled siblings (HH)	-0.0001	-0.0001
	(0.002)	(0.002)
Fit statistics		
Bartik F-stat (1st stage)	502.45	502.45
Observations	13,518	13,518
R^2	0.25499	0.25499
Mean of dep. var.	0.879	0.879

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 21: Dropout Reasons, IV Estimates (Lag 2)

Dependent Variables:	Dropout: financial	Dropout: family	Dropout: no interest	Dropout: pregnancy
Model:	(1)	(2)	(3)	(4)
Variables				
Tourism exp.,	0.019	0.074	0.137	-0.102
lag 2 (100M USD)	(0.050)	(0.067)	(0.151)	(0.099)
Female	-0.031***	-0.004	-0.039***	0.079***
	(0.011)	(0.009)	(0.010)	(0.011)
Age	-0.003	-0.005**	-0.005	-0.003
	(0.004)	(0.002)	(0.003)	(0.002)
No. enrolled siblings (HH)	0.008	-0.003	-0.003	0.009*
	(0.006)	(0.003)	(0.004)	(0.005)
Fit statistics				
Bartik F-stat (1st stage)	67.093	67.093	67.093	67.093
Observations	1,637	1,637	1,637	1,637
R^2	0.08795	0.03729	0.01672	0.05894
Mean of dep. var.	0.038	0.020	0.043	0.044

Notes: Sample: children ages 5–19 not currently enrolled. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 22: School Attendance, IV Estimates (Lag 2)

Dependent Variables:	Days sent per week		Kept from school (any)
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.107	-0.107	0.100*
lag 2 (100M USD)	(1.51)	(1.51)	(0.058)
Female	0.035	0.035	0.0008
	(0.113)	(0.113)	(0.005)
Age	-0.029***	-0.029***	-0.007***
	(0.011)	(0.011)	(0.0008)
No. enrolled siblings (HH)	-0.035	-0.035	-0.006
	(0.078)	(0.078)	(0.004)
Fit statistics			
Bartik F-stat (1st stage)	434.51	434.51	439.36
Observations	11,412	11,412	11,881
R^2	0.03447	0.03447	0.22638
Mean of dep. var.	18.803	18.803	0.139

Notes: Sample: currently enrolled children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 23: School Absence Reasons, IV Estimates (Lag 2)

Dependent Variables:	Kept out, any reason	Kept out, money	Kept out, transport	Kept out, illness	Kept out, work
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	0.129	-0.154	-0.004	0.369	-0.0008
lag 2 (100M USD)	(0.211)	(0.105)	(0.021)	(0.319)	(0.002)
Female	-0.026***	-0.016**	-5.49×10^{-5}	-0.003	-0.0004
	(0.006)	(0.007)	(0.001)	(0.006)	(0.0004)
Age	-0.002	0.002	0.0004***	-0.004***	3.56×10^{-5}
	(0.001)	(0.0009)	(0.0001)	(0.0006)	(5.29×10^{-5})
No. enrolled siblings (HH)	0.015***	0.022***	0.0005	-0.004	-8.92×10^{-5}
	(0.004)	(0.005)	(0.0006)	(0.003)	(0.0002)
Fit statistics					
Bartik F-stat (1st stage)	439.36	439.36	439.36	439.36	439.36
Observations	11,881	11,881	11,881	11,881	11,881
R^2	0.13273	0.06072	0.01883	-0.04711	0.00632
Mean of dep. var.	0.188	0.108	0.003	0.063	0.001

Notes: Sample: enrolled children; absence measured by annual recall (occasional or frequent). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 24: Transport Absence by Urban/Rural, IV Estimates (Lag 2)

Dependent Variable:		Kept out, transport	
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.004	-6.38×10^{-5}	0.007
lag 2 (100M USD)	(0.021)	(0.019)	(0.009)
Female	-5.49×10^{-5}	-0.001	0.001
	(0.001)	(0.002)	(0.002)
Age	0.0004***	0.0005**	0.0003*
	(0.0001)	(0.0002)	(0.0002)
No. enrolled siblings (HH)	0.0005	0.001	5.08×10^{-5}
	(0.0006)	(0.001)	(0.0007)
Fit statistics			
Bartik F-stat (1st stage)	439.36	110.99	655.04
Observations	11,881	4,501	7,380
R^2	0.01883	0.01446	0.03019
Mean of dep. var.	0.003	0.002	0.004

Notes: Outcome is an indicator equal to 1 if the child was kept from school occasionally or frequently due to transportation costs in the past year. Columns (2) and (3) restrict the sample to urban and rural enrolled children, respectively. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 25: School Expenditure by Category, IV Estimates (Lag 2)

Dependent Variables:	Tuition (USD)	Books (USD)	Transport (USD)	Uniform (USD)	Lunch and snacks (USD)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp.,	25.3	-9.77	-33.2	13.6	108.6
lag 2 (100M USD)	(40.0)	(11.2)	(68.3)	(23.2)	(89.4)
Female	-4.42	5.63***	25.7***	-8.40***	15.0**
	(6.76)	(1.87)	(7.99)	(1.77)	(6.54)
Age	-7.29***	1.15***	31.9***	2.69***	29.3***
	(0.770)	(0.247)	(2.48)	(0.262)	(1.30)
No. enrolled siblings (HH)	-12.8***	-9.34***	-11.7***	-8.39***	-28.2***
	(2.98)	(0.951)	(2.64)	(1.05)	(3.41)
Fit statistics					
Bartik F-stat (1st stage)	990.13	808.88	771.86	808.75	802.52
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.04441	0.09251	0.08191	0.05555	0.11900
Mean of dep. var.	48.388	70.385	258.767	88.556	464.417

Notes: Sample: enrolled children; real 2024 USD; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 26: School Expenditure by Category (Log), IV Estimates (Lag 2)

Dependent Variables:	Log tuition	Any books expenses	Log transport	Log uniform	Log lunch and snacks
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (100M USD)	-0.137			0.215	-0.842
	(1.27)			(1.41)	(1.41)
Female	-0.010	0.024***	0.991***	0.053	-0.073
	(0.146)	(0.008)	(0.219)	(0.136)	(0.109)
Age	-0.552***	-0.014***	1.13***	0.025	0.183***
	(0.036)	(0.002)	(0.039)	(0.019)	(0.020)
No. enrolled siblings (HH)	-0.394***	-0.031***	-0.372***	-0.246**	-0.200***
	(0.084)	(0.005)	(0.105)	(0.098)	(0.058)
Tourism exp., lag 1 (100M USD)		-0.157*	2.29		
		(0.084)	(2.18)		
Fit statistics					
Bartik F-stat (1st stage)	990.13			808.75	802.52
Bartik F-stat (1st stage)		634.72	638.01		
Observations	9,246	10,281	10,283	10,304	10,337
R^2	0.08912	0.06817	0.13568	0.02251	0.03080
Mean of dep. var.	-17.676	0.811	-2.766	2.586	4.455

Notes: Sample: enrolled children; 2014 excluded due to survey design. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 27: Examination Attainment, IV Estimates (Lag 2)

Dependent Variables:	GNAT	CXC Basic	CXC General	CXC: math and English	CAPE/ A-Level	Degree
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Tourism exp.,	0.0002	-0.0001	-0.021	-0.024	0.051	-0.004
lag 2 (100M USD)	(0.037)	(0.033)	(0.037)	(0.020)	(0.091)	(0.020)
Female	0.031***	0.030***	0.025***	0.011***	0.019***	0.004*
	(0.004)	(0.004)	(0.004)	(0.002)	(0.004)	(0.002)
Age	0.021***	0.019***	0.016***	0.007***	0.010***	0.003***
	(0.0008)	(0.0009)	(0.0010)	(0.0008)	(0.001)	(0.0007)
No. enrolled siblings (HH)	-0.005**	-0.006**	-0.006**	-0.002**	-0.008***	-0.002**
	(0.002)	(0.002)	(0.002)	(0.0009)	(0.002)	(0.0007)
Fit statistics						
Bartik F-stat (1st stage)	502.45	502.45	502.45	502.45	229.12	229.12
Observations	13,518	13,518	13,518	13,518	6,209	6,209
R^2	0.14588	0.13853	0.11850	0.05207	0.03914	0.02344
Sample						
School-age (5–19)	Yes	Yes	Yes	Yes	No	No
Upper secondary (16–25)	No	No	No	No	Yes	Yes
Mean of dep. var.	0.061	0.056	0.046	0.020	0.041	0.006

Notes: Sample varies by outcome: ages 5–19 for GNAT through CXC qualifications; ages 16–25 for A-Level and Degree. Each outcome is a binary indicator equal to 1 if the respondent has attained at least that qualification level (or higher). All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 28: Upper Secondary and Tertiary Attainment, IV Estimates (Lag 2)

Dependent Variables:	At least CAPE/A-Level	At least degree		Currently in college	
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Tourism exp., lag 2 (100M USD)	-0.029	-0.013	-0.044	-0.067	0.024
	(0.045)	(0.011)	(0.032)	(0.063)	(0.091)
Female	0.029***	0.009***	0.036***	0.038***	0.032*
	(0.005)	(0.002)	(0.006)	(0.006)	(0.018)
Age	0.010***	0.004***	-0.0004	0.001	-0.0003
	(0.002)	(0.0008)	(0.0008)	(0.0009)	(0.002)
No. enrolled siblings (HH)				-0.006**	
				(0.003)	
Fit statistics					
Bartik F-stat (1st stage)	296.30	296.30	296.30	229.12	10.237
Observations	7,799	7,799	7,799	6,209	304
R^2	0.05034	0.02775	0.03256	0.03247	0.26421
Mean of dep. var.	0.049	0.010	0.067	0.065	0.030

Notes: Sample: ages 16–25. All specifications include development area and year fixed effects; standard errors clustered by development area.

Table 29: Skills Program Enrollment and Completion, IV Estimates (Lag 2)

Dependent Variables:	Enrolled in skills program		Skills program diploma
Model:	(1)	(2)	(3)
Variables			
Tourism exp.,	-0.029	-0.029	-0.246
lag 2 (100M USD)	(0.120)	(0.120)	(0.302)
Female	0.003	0.003	-0.005
	(0.002)	(0.002)	(0.012)
Age	-0.0005	-0.0005	0.017***
	(0.001)	(0.001)	(0.004)
No. enrolled siblings (HH)	-0.001**	-0.001**	-0.009
	(0.0005)	(0.0005)	(0.005)
Fit statistics			
Bartik F-stat (1st stage)	9.2371	9.2371	14.643
Observations	1,576	1,576	1,866
R^2	0.13116	0.13116	0.11363
Mean of dep. var.	0.003	0.003	0.077

Notes: Sample: children ages 5–19. All specifications include development area and year fixed effects; standard errors clustered by development area.

7.6 Coefficient Plots: One-Year Lag Specification

Figures 6–15 display coefficient plots for all main regression outcomes under the primary one-year lagged specification. Each figure plots the tourism expenditure IV coefficient with 95% (wide) and 90% (narrow) confidence intervals. Columns correspond to the specifications shown in the numbered tables above.

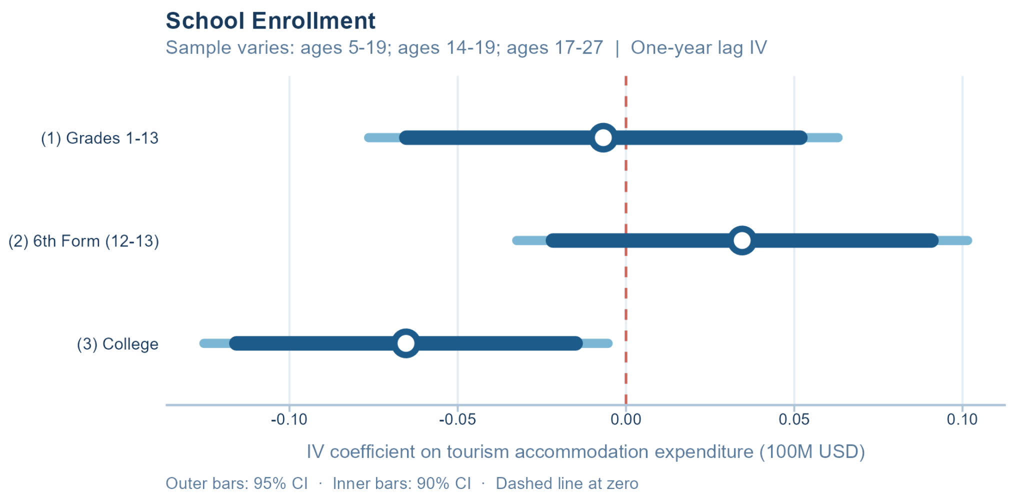

Figure 6: Enrollment: IV Coefficient Estimates (Lag 1)

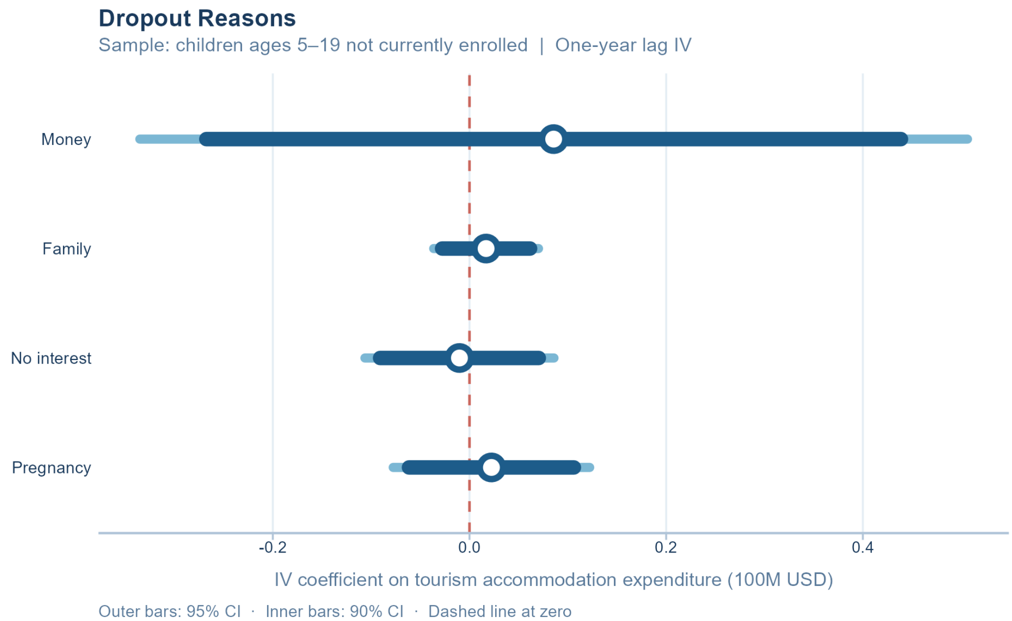

Figure 7: Dropout Reasons: IV Coefficient Estimates (Lag 1)

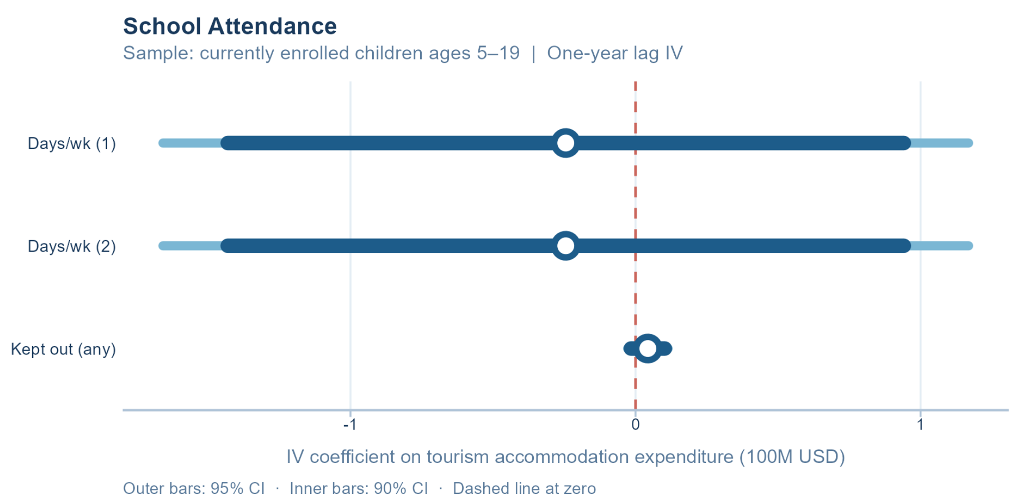

Figure 8: Attendance Levels: IV Coefficient Estimates (Lag 1)

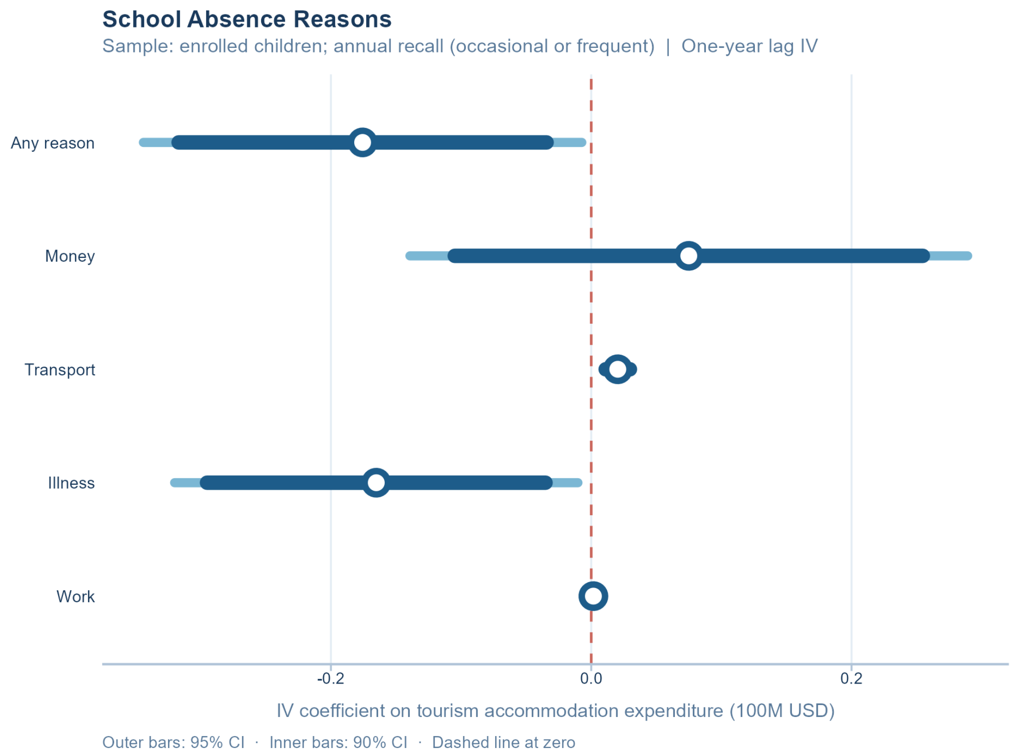

Figure 9: Absence Reasons: IV Coefficient Estimates (Lag 1)

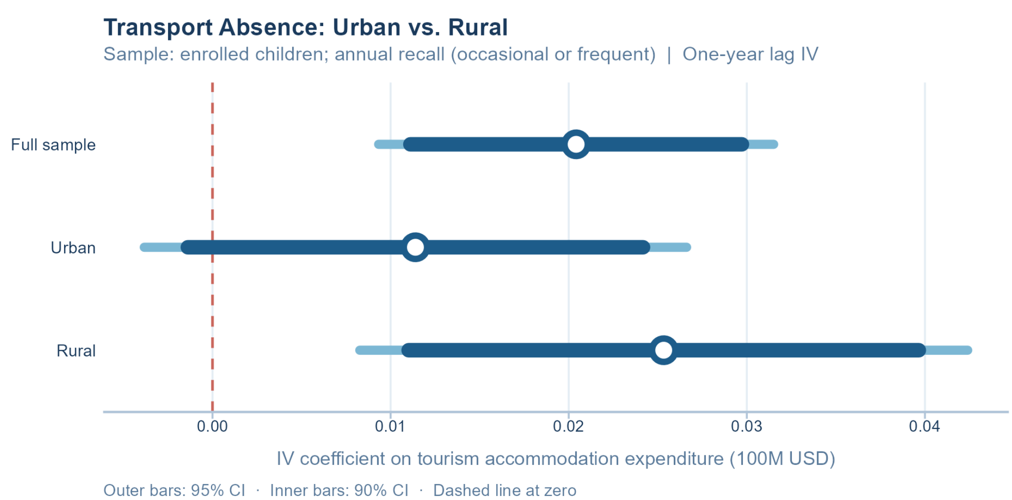

Figure 10: Transport Absence, Urban vs. Rural: IV Coefficient Estimates (Lag 1)

Figure 11: Skills Program: IV Coefficient Estimates (Lag 1)

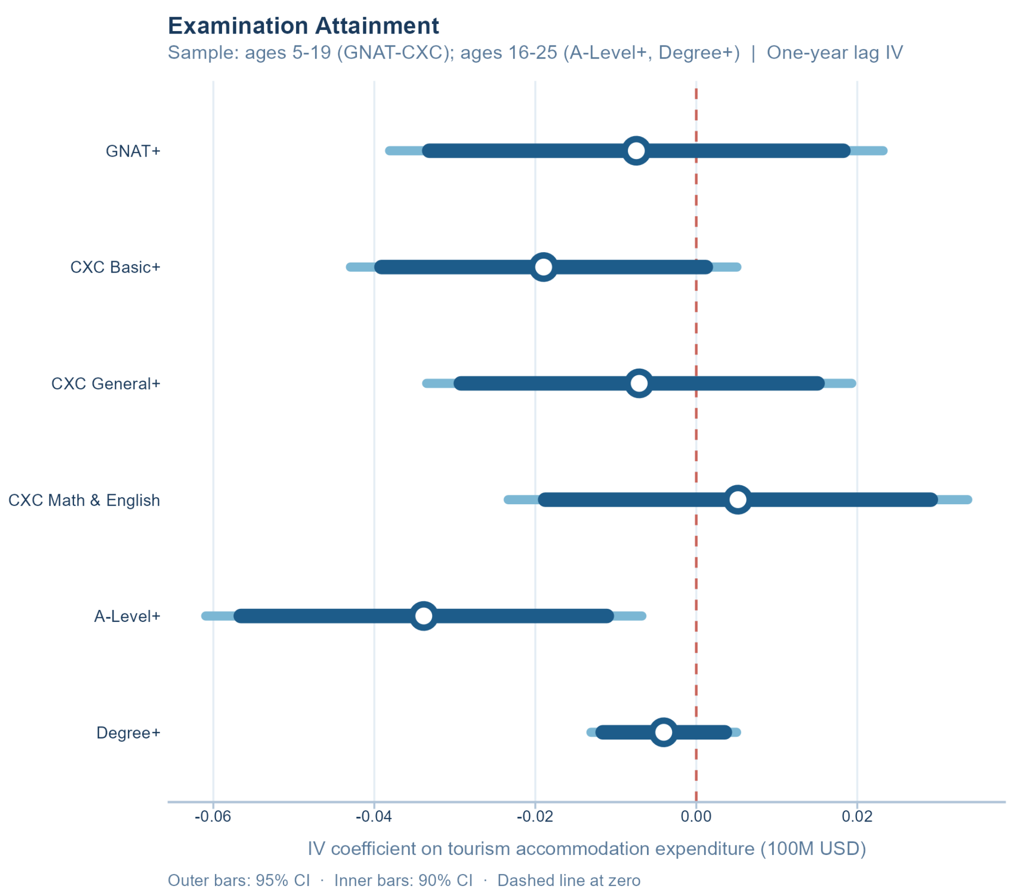

Figure 12: Examination Attainment: IV Coefficient Estimates (Lag 1)

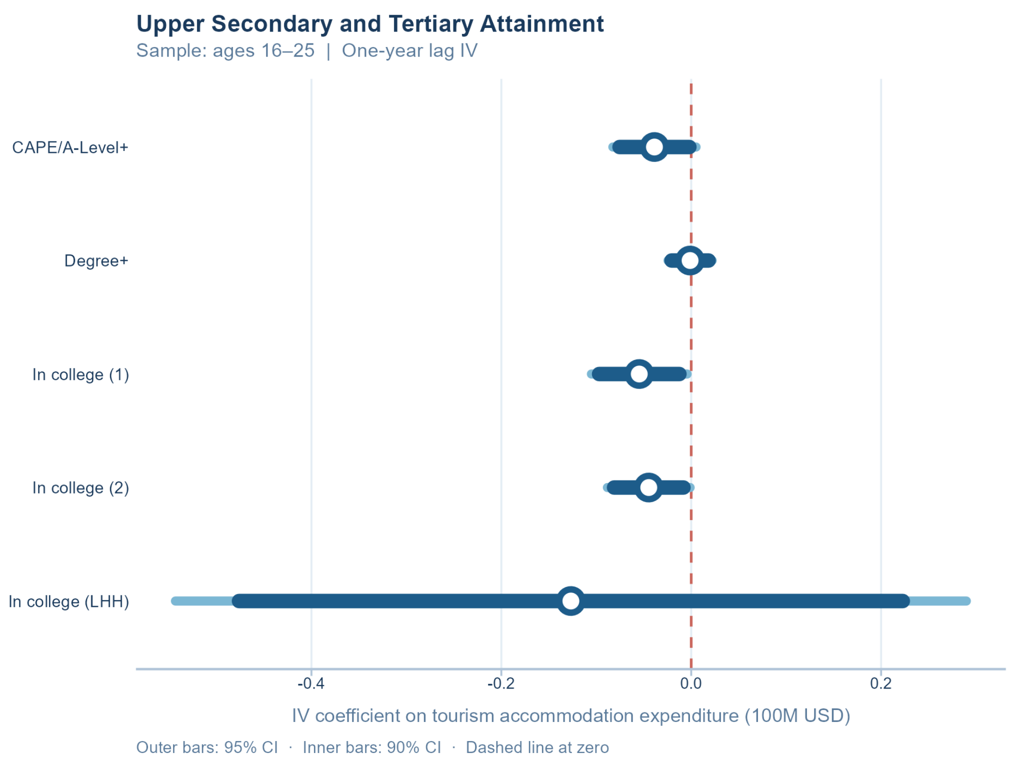

Figure 13: Tertiary Attainment: IV Coefficient Estimates (Lag 1)

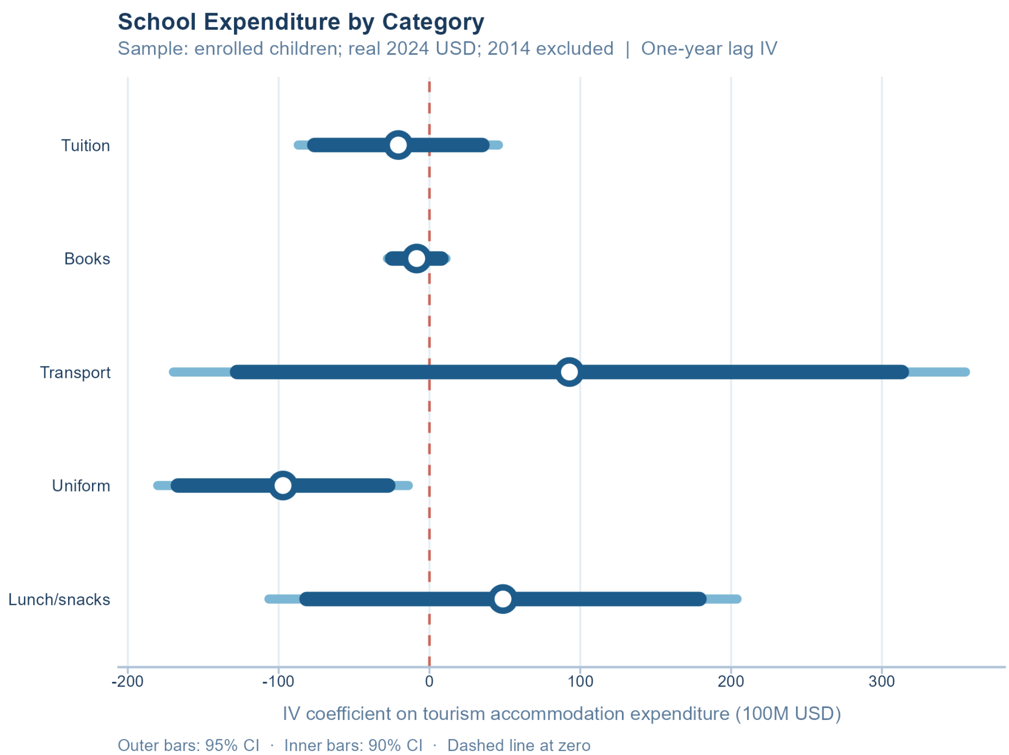

Figure 14: Expenditure Levels: IV Coefficient Estimates (Lag 1)

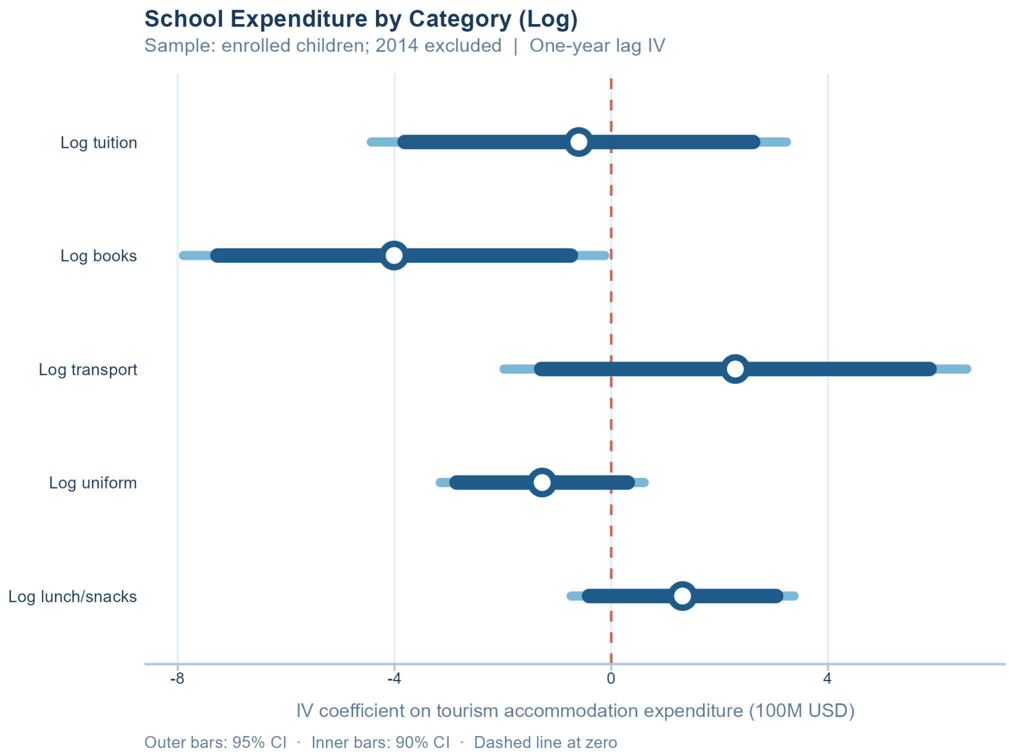

Figure 15: Log Expenditures: IV Coefficient Estimates (Lag 1)

7.7 Coefficient Plots: Contemporaneous Specification

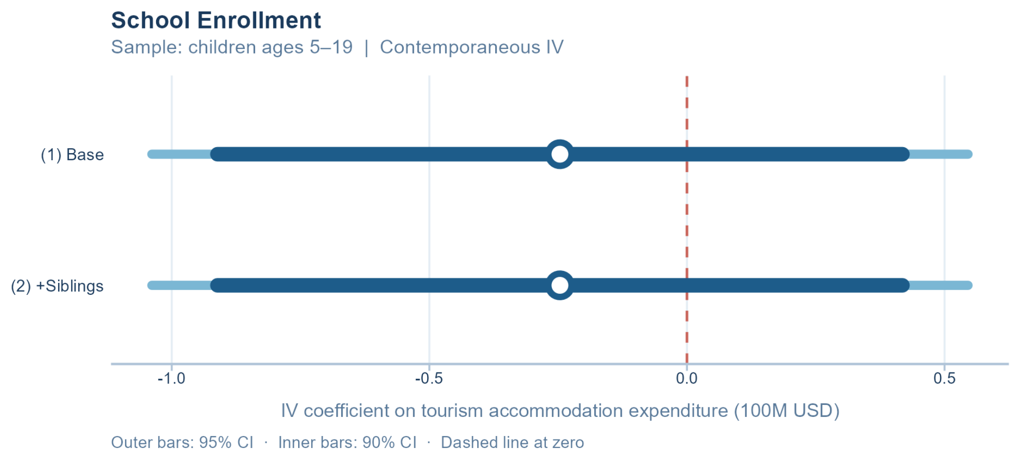

Figure 16: Enrollment: IV Coefficient Estimates (Contemporaneous)

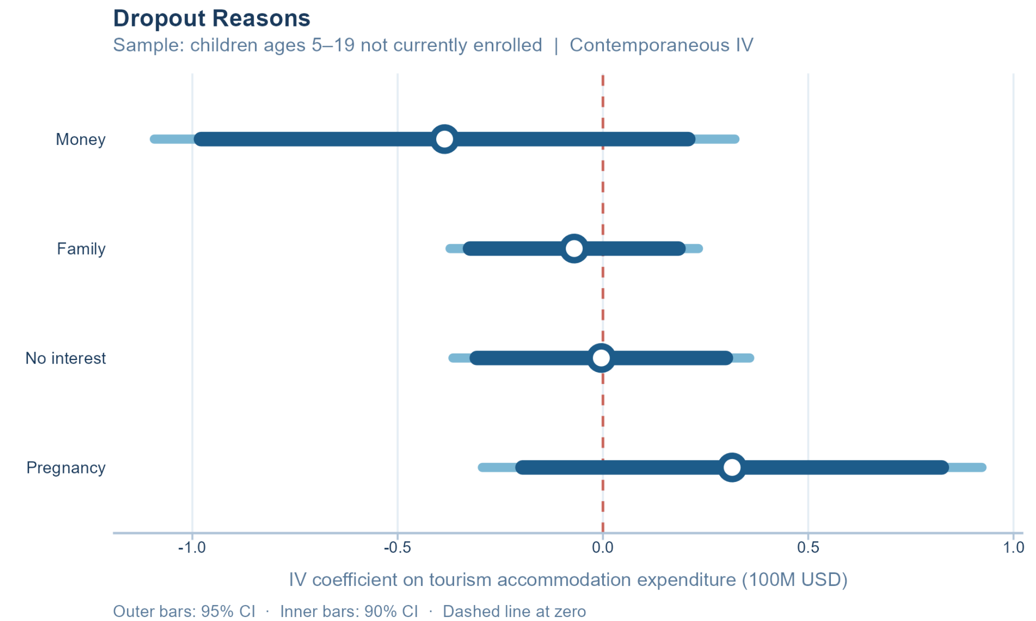

Figure 17: Dropout Reasons: IV Coefficient Estimates (Contemporaneous)

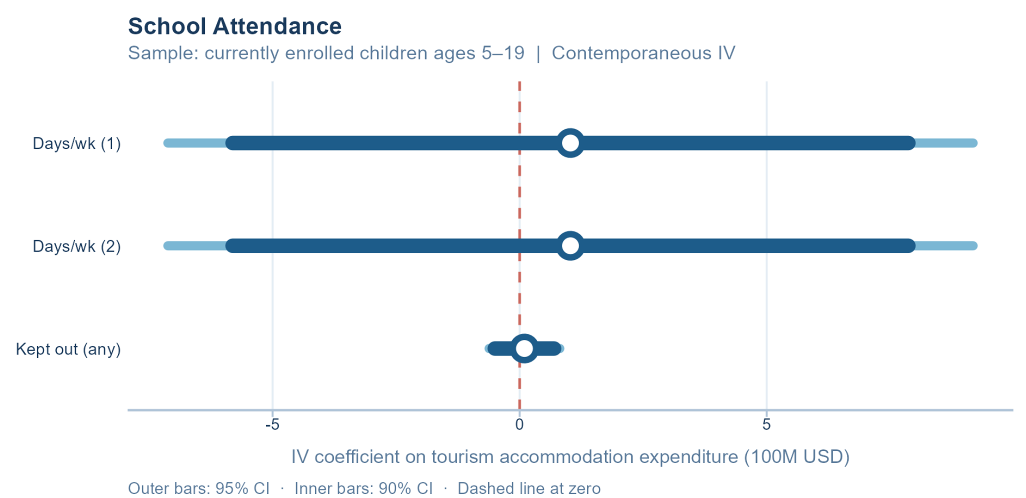

Figure 18: Attendance Levels: IV Coefficient Estimates (Contemporaneous)

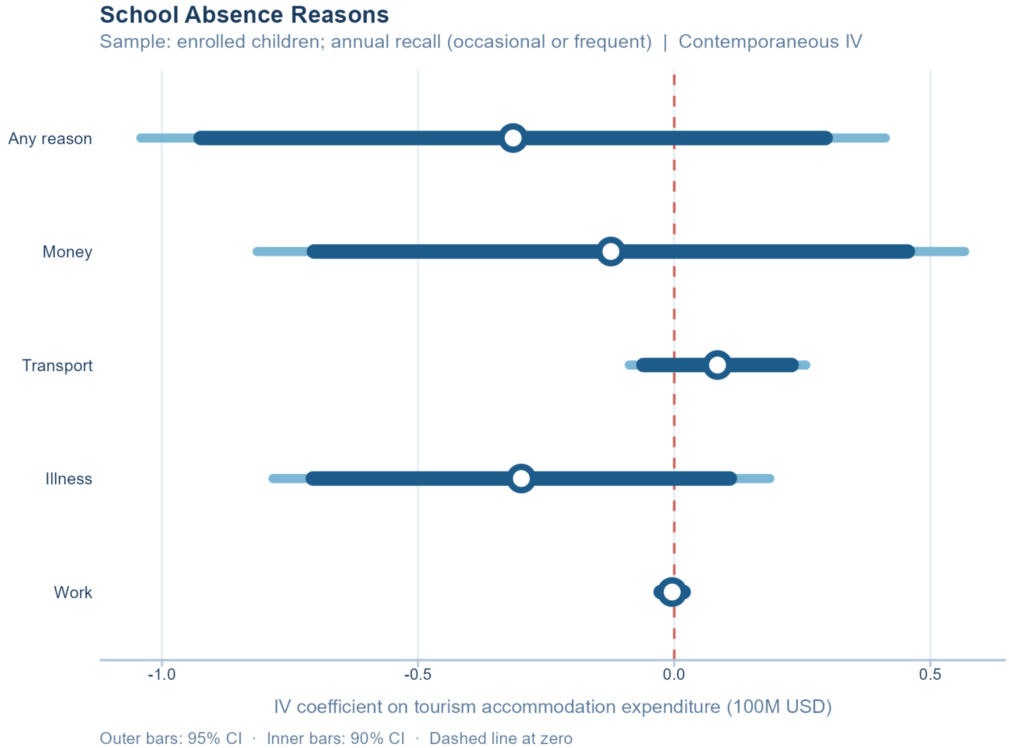

Figure 19: Absence Reasons: IV Coefficient Estimates (Contemporaneous)

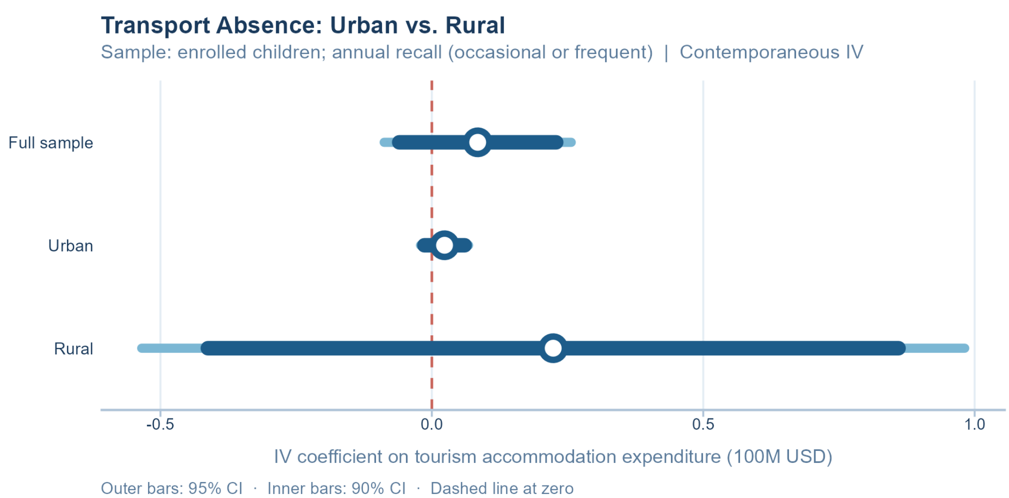

Figure 20: Transport Absence, Urban vs. Rural: IV Coefficient Estimates (Contemporaneous)

Figure 21: Skills Program: IV Coefficient Estimates (Contemporaneous)

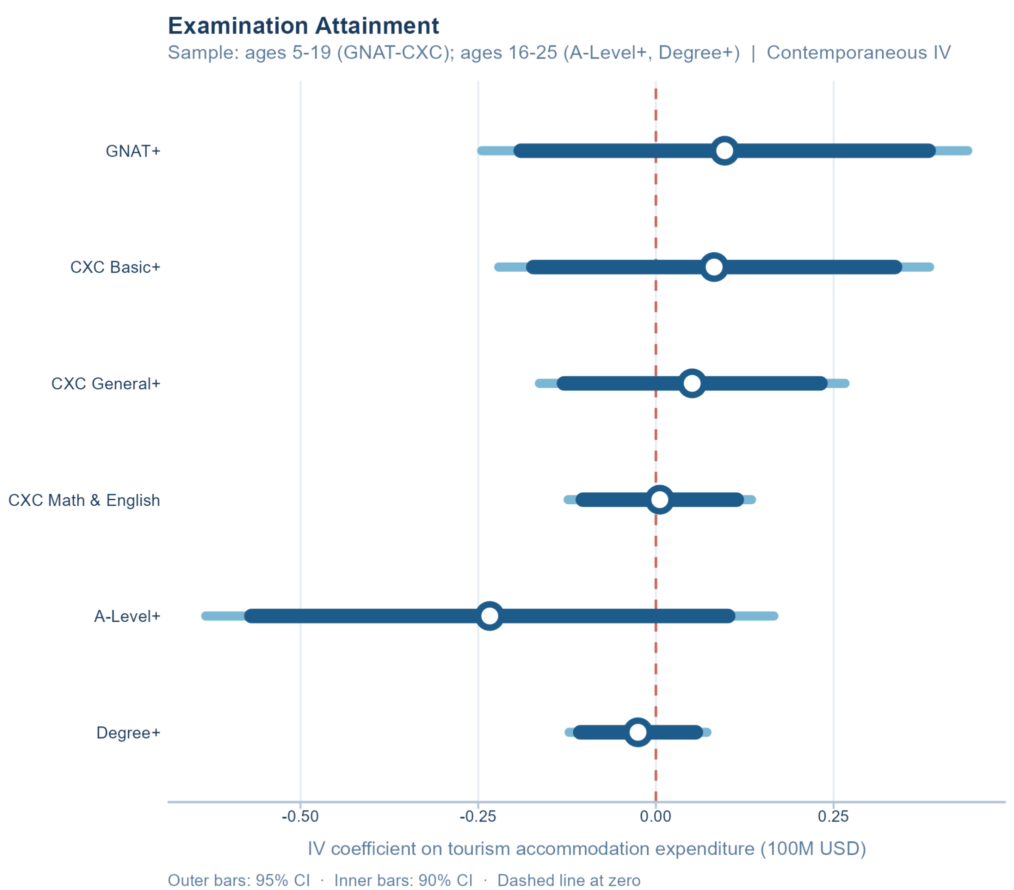

Figure 22: Examination Attainment: IV Coefficient Estimates (Contemporaneous)

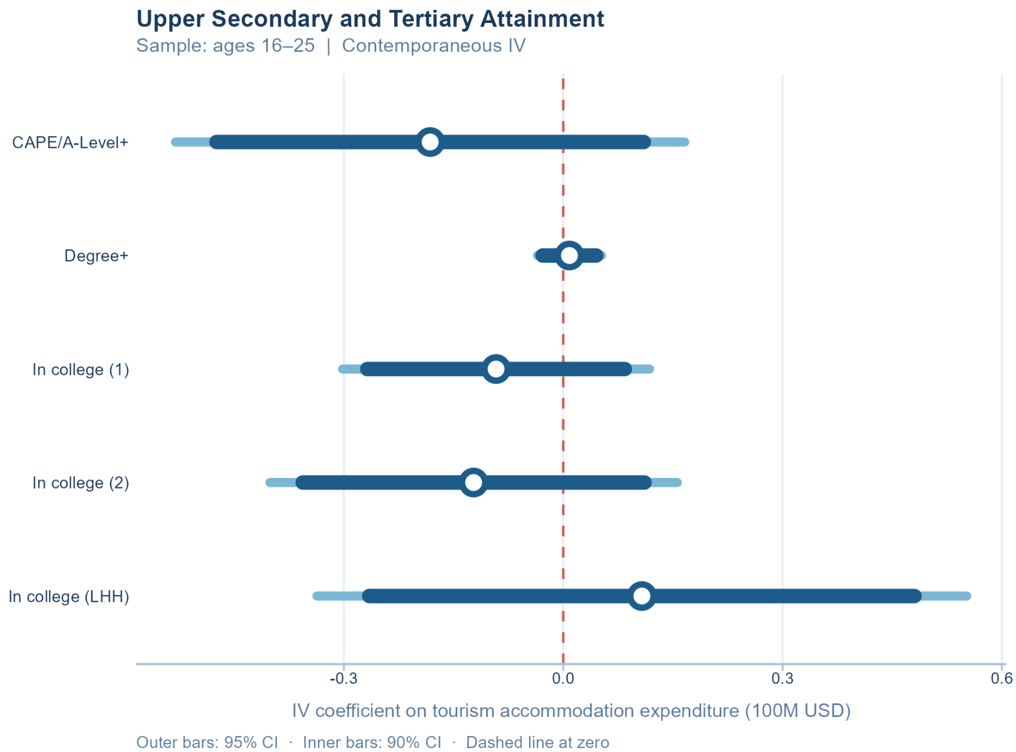

Figure 23: Tertiary Attainment: IV Coefficient Estimates (Contemporaneous)

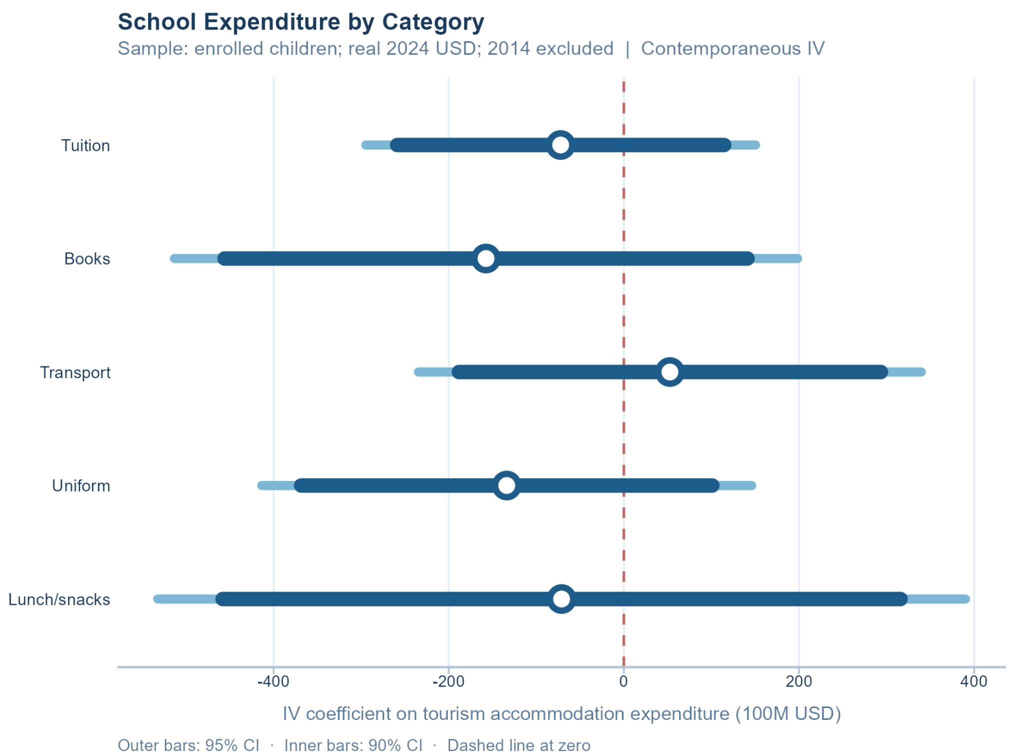

Figure 24: Expenditure Levels: IV Coefficient Estimates (Contemporaneous)

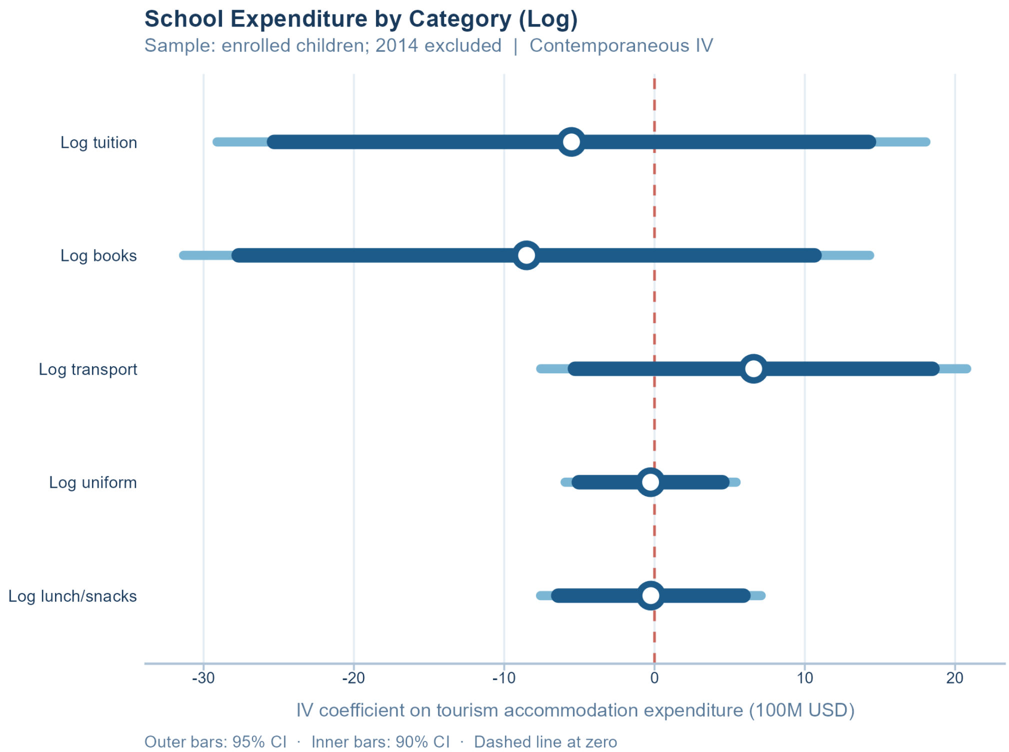

Figure 25: Log Expenditures: IV Coefficient Estimates (Contemporaneous)

7.8 Coefficient Plots: Two-Year Lag Specification

Figure 26: Enrollment: IV Coefficient Estimates (Lag 2)

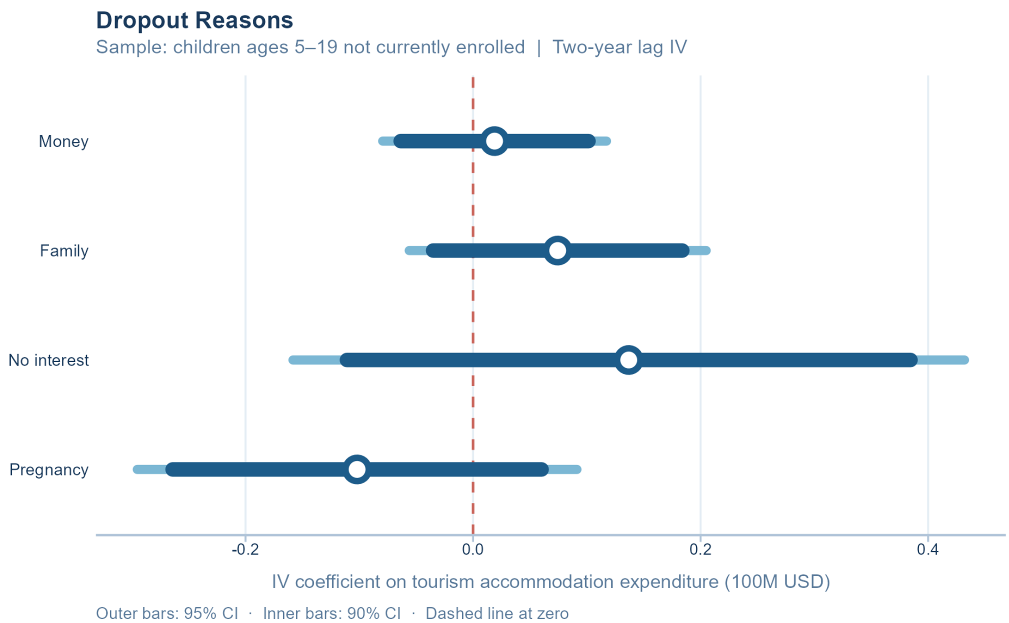

Figure 27: Dropout Reasons: IV Coefficient Estimates (Lag 2)

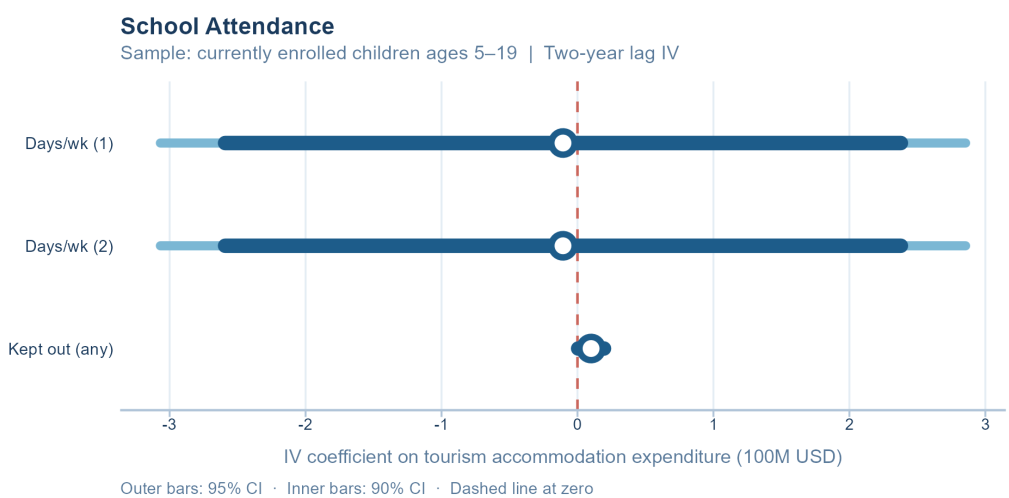

Figure 28: Attendance Levels: IV Coefficient Estimates (Lag 2)

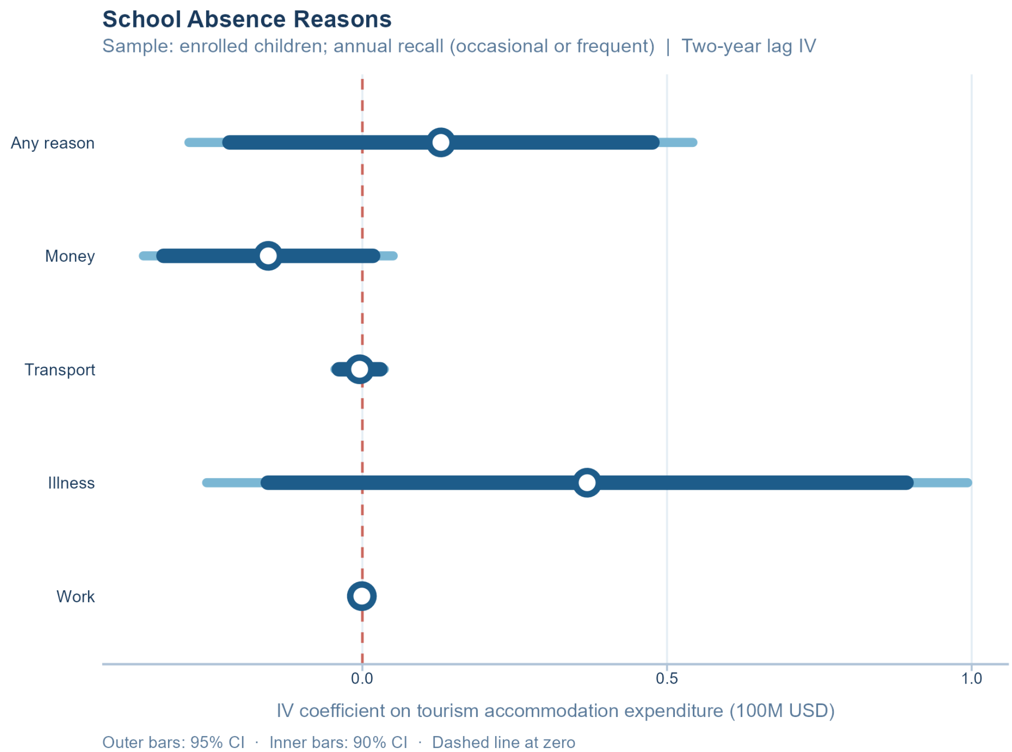

Figure 29: Absence Reasons: IV Coefficient Estimates (Lag 2)

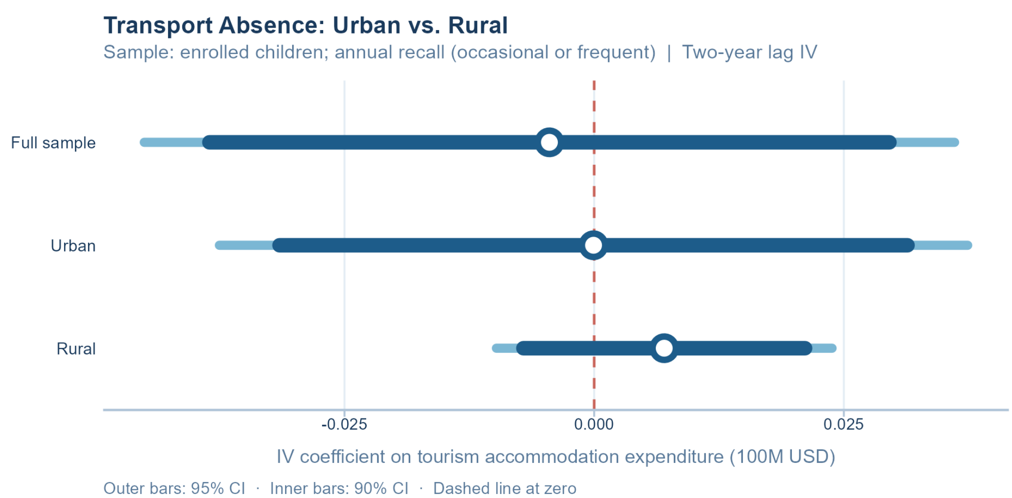

Figure 30: Transport Absence, Urban vs. Rural: IV Coefficient Estimates (Lag 2)

Figure 31: Skills Program: IV Coefficient Estimates (Lag 2)

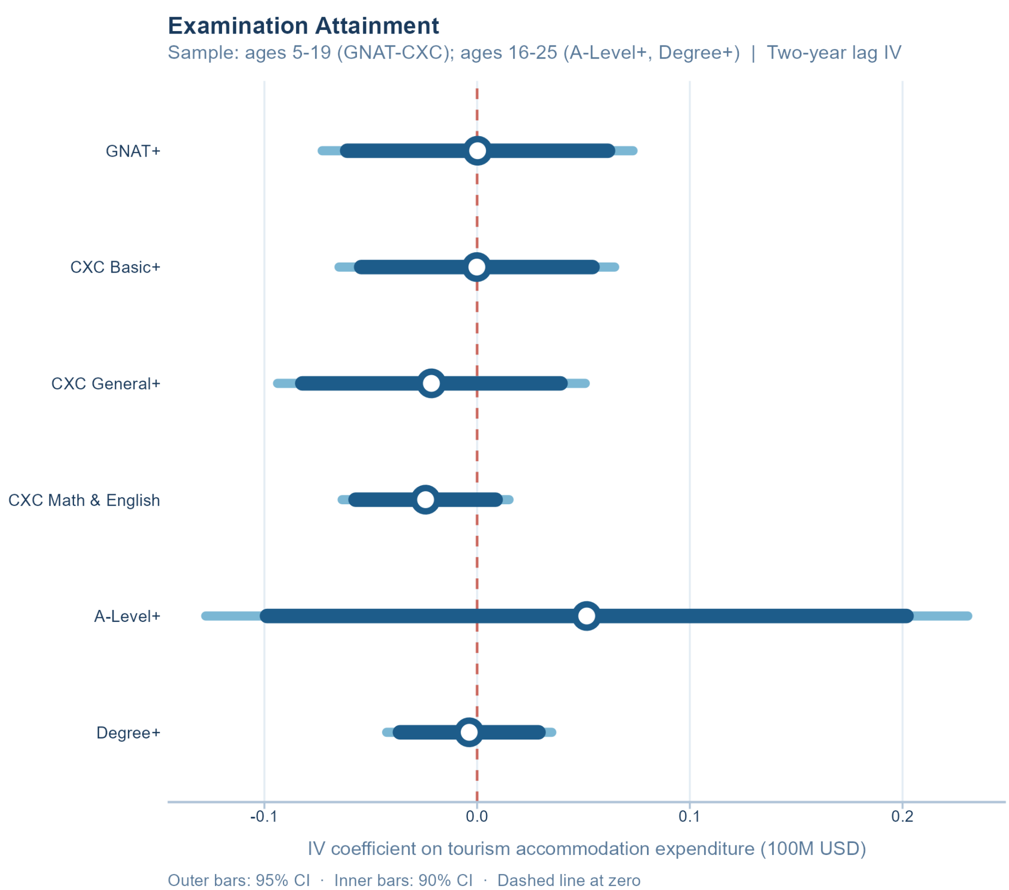

Figure 32: Examination Attainment: IV Coefficient Estimates (Lag 2)

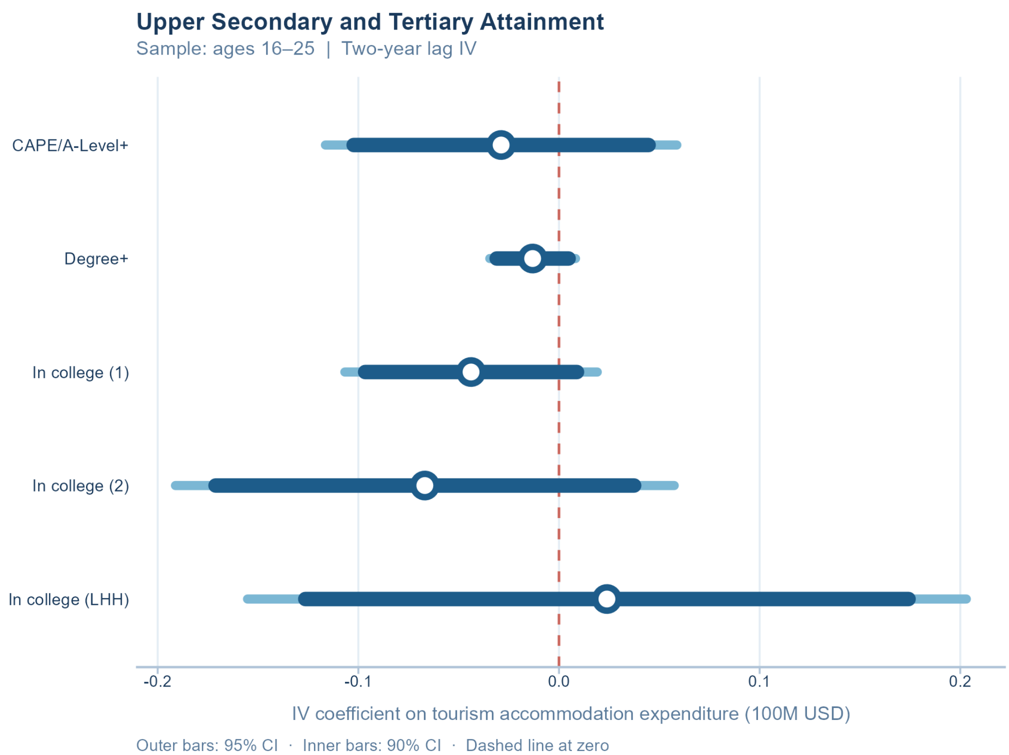

Figure 33: Tertiary Attainment: IV Coefficient Estimates (Lag 2)

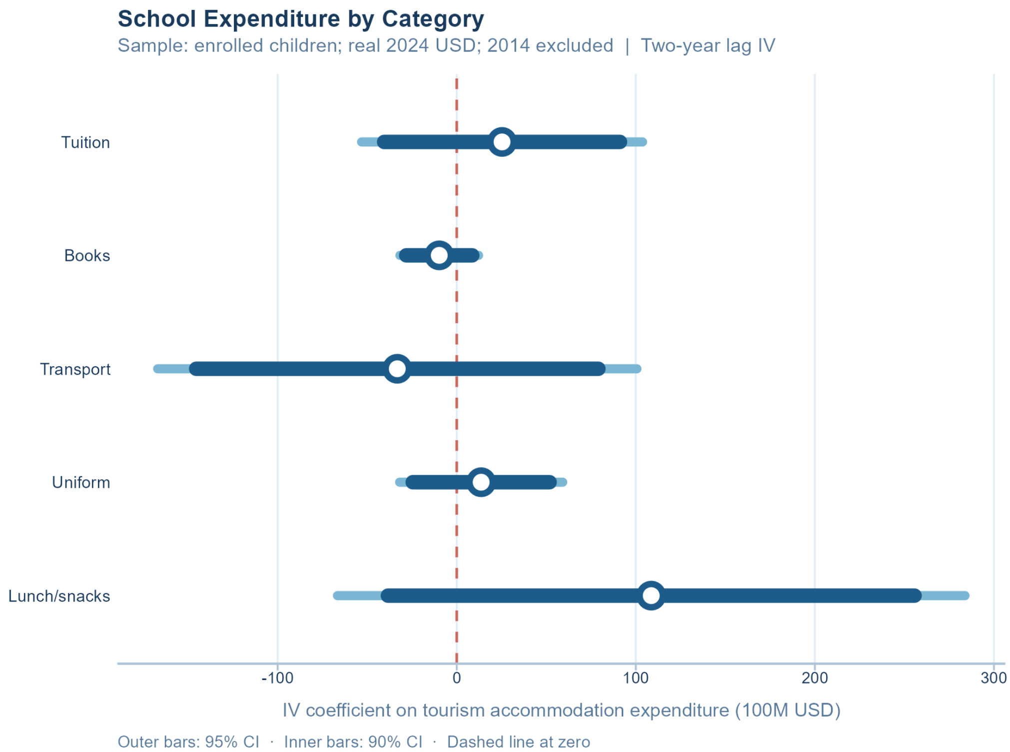

Figure 34: Expenditure Levels: IV Coefficient Estimates (Lag 2)

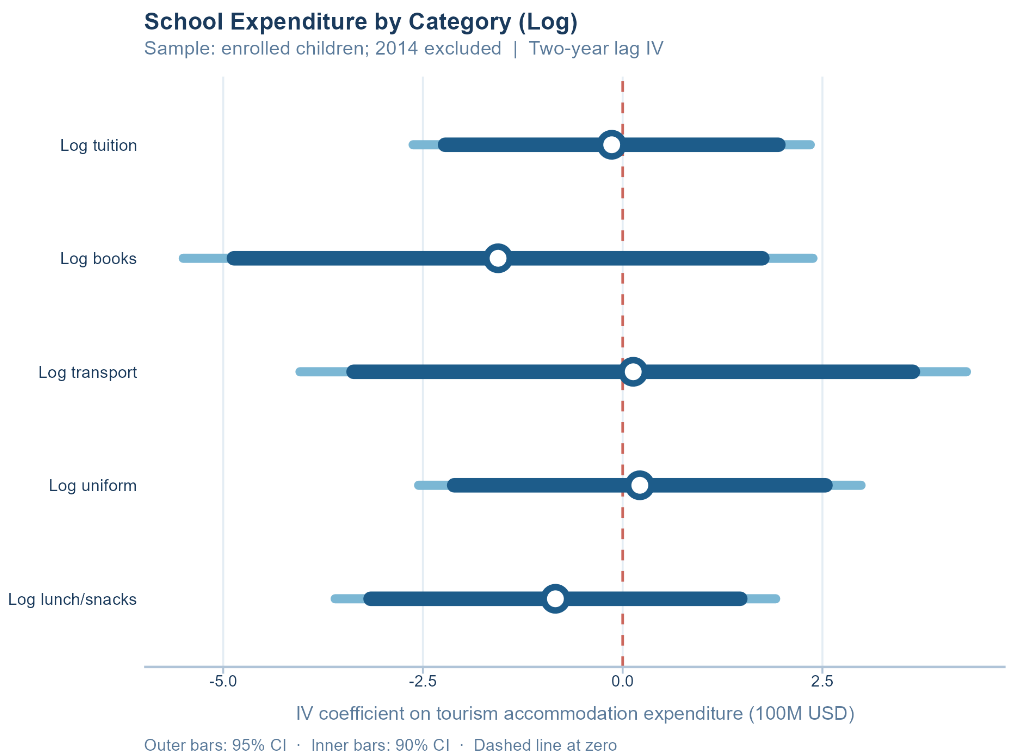

Figure 35: Log Expenditures: IV Coefficient Estimates (Lag 2)

7.9 Tourism Summary Statistics

[Appendix placeholder.]

7.10 Additional Household and Education Summary Statistics

[Appendix placeholder.]